Two Hidden Layers Suffice: The Caterpillar Net

Jakob Paul Zimmermann
FU Berlin, TU Berlin
jakob.paul.zimmermann@campus.tu-berlin.de
https://jakob-paul-zimmermann.com

Abstract

Every continuous piecewise linear (CPWL) function is a finite signed sum of maxima of affine functions, making maximum representation a central problem in the study of the expressiveness of ReLU networks. Earlier work disproved the conjectured base-two depth bound, leaving open whether a constant number of hidden layers suffices to represent the maximum. Independently and concurrently, Brunck, Hertrich, Rueß, Stade, and Yehudayoff established that two hidden layers suffice for all CPWL functions. We give the *Caterpillar Net*, a non-recursive explicit construction representing the maximum of N real inputs exactly with two hidden layers of widths $\binom{N}{2} + N + 1$ and $2^{O(N \log N)}$, rational weights, and zero biases. Every expansion of the maximum in its finite vocabulary compiles with the same first hidden layer. Sharing its first hidden layer across a Wang–Sun representation gives two-hidden-layer networks for all CPWL functions. [1]

1 Introduction

In this paper we investigate the representation capabilities of ReLU networks: what functions are representable by such networks of different depths and widths? ReLU networks represent precisely the continuous piecewise-linear (CPWL) functions with finitely many pieces [ABMM18]. A sharp question is therefore: given a CPWL function, what is the smallest depth, or the smallest width at a prescribed depth, needed to represent it exactly [HBD+23, Saf26]? The maximum is a universal building block for such functions: every CPWL function is a finite signed sum of maxima of affine functions [WS05]. Write

$$\max_N(x) := \max_{0 \le i < N} x_i. \tag{1}$$

Theorem (Wang–Sun, [WS05], `Lean_1`)**.** *Let $d \ge 1$ and let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous and affine on each member of a finite polyhedral cover. There are $R \in \mathbb{N}$, affine maps and signs*

$$A_t : \mathbb{R}^d \longrightarrow \mathbb{R}^{d+1}, \qquad \varepsilon_t \in \{-1, 1\} \qquad (1 \le t \le R) \tag{2}$$

such that

$$f(x) = \sum_{t=1}^{R} \varepsilon_t \max_{d+1}(A_t(x)) \qquad \textit{for every } x \in \mathbb{R}^d. \tag{3}$$

[1] Our results were obtained over months of proof search in Lean using Claude Code and Codex; the AI methodology statement describes their roles. The Lean code is available in the accompanying repository https://github.com/JayPiZimmermann/Max_Caterpillar. Theorem titles link to the Lean formalization; clickable equation numbers link to supporting declarations or written proofs.

Thus a two-hidden-layer representation of $\max_{d+1}$ gives one for every CPWL function on $\mathbb{R}^d$: precompose with the affine maps and add the networks in parallel. Section 7 reuses their first hidden layer; the representation theorems and the resulting networks are formalized.

Our contribution. The *Caterpillar Net* gives a nonrecursive two-hidden-layer representation of the maximum with explicit bounds on both hidden-layer widths. The construction uses a finite dictionary of support functions of caterpillar polytopes and their faces. The maximum is the support function of a simplex. We first express this support function as an alternating sum of caterpillar support functions, up to a linear combination of support functions of lower-dimensional polytopes. We then represent the remaining error using faces of the same caterpillar polytopes and elementary ReLU features. Normalizing and merging these descriptions yields a width bound for the complete network, including all correction terms.

Write $\sigma(t) := \max(0, t)$ applied coordinatewise to vectors. A network has an input layer of N coordinates, first and second hidden layers of widths h_1 and h_2, respectively, and a scalar output layer. With real weight matrices W_1, W_2, output vector w_3, and biases b_1, b_2, b_3 of the corresponding dimensions, it evaluates as

$$x \longmapsto w_3^\top \sigma\big(W_2\sigma(W_1x + b_1) + b_2\big) + b_3. \tag{4}$$

There are no skip connections; equality is required on all of $\mathbb{R}^N$. A network has *rational weights* if every entry of W_1, W_2, and w_3 is rational, and *zero biases* if b_1, b_2, and b_3 vanish.

Theorem 1 (Caterpillar Net, `Lean_2`, Proof)**.** *For every $N \geq 3$, there is a two-hidden-layer ReLU network with rational weights and zero biases computing $\max_N$ with widths*

$$h_1 = \binom{N}{2} + N + 1, \qquad h_2 = 3^{N-2}C_N - N(N-1) + 2, \tag{5}$$

where

$$C_N := \sum_{m=2}^{N-1} \binom{N}{m} \frac{m!}{2}(N-m)m^{N-m-1}. \tag{6}$$

Thus $h_1 = O(N^2)$ and $h_2 = 2^{O(N \log N)}$ as $N \to \infty$. The rational coefficients of this construction are specified by the finite linear system in Equation (21)*.*

Further results. The construction chooses one basis of the caterpillar dictionary extending a basis of its auxiliary span, determines its coefficients by rational evaluations, and compiles them into the network in Section 3.2. The basis has q_N nonauxiliary functions, so this same network has $q_N + 2$ neurons in its second hidden layer before zero padding to the stated width. Conversely, Section 4 gives exponential lower bounds for caterpillar expansions and proves $q_N \geq (N-2)!$. Section 7 applies the construction to sums of maxima.

2 Related work

The conjecture that $\max\{0, x_1, \ldots, x_{2^k}\}$ requires more than k hidden layers [HBD+23] was disproved for $k = 2$ by the maximum of five inputs in [BBH+26]. Exact computer-assisted searches subsequently obtained two-hidden-layer representations up to $N = 12$ in [RAB+26] and, independently, up to $N = 8$ in [WB26]. The improved general depth bounds remained logarithmic, leaving constant depth open.

Setting all biases to zero does not change a network that computes a positively homogeneous function such as $\max_N$ [HBD+23, Proposition 2.3], [HHL23, Lemma 7], so biases give no additional power for

representing the maximum. Depth lower bounds hold for integer weights [HHL23] and rational weights whose denominators are restricted to powers of a fixed integer [AHM25]. Our weights are rational, but their denominators are not restricted in this way, so neither bound applies. Other results concern braid-fan-compatible networks [GHL25] and first hidden layer width bounds [Saf26, Lu26].

Theorem 2 (Brunck, Hertrich, Rueß, Stade, and Yehudayoff, [BHR+26])**.** *Every continuous piecewise affine function $f : \mathbb{R}^d \to \mathbb{R}$ with finitely many affine pieces can be represented exactly by a ReLU network with two hidden layers.*

Their first proof proceeds by induction on dimension, using homological algebra to construct a linear combination of support functions of pyramids over zonotopes that agrees with the simplex support function modulo lower-dimensional terms. Our construction restricts these building blocks to an explicitly specified finite family, the caterpillar polytopes; we deliver a geometric proof of the corresponding identity.

Independently and concurrently, they developed a second proof based on combinatorial cancellation, with building blocks closely related to ours. Appendix D identifies them exactly: for nonempty marked sets their pyramids are translated caterpillar polytopes, and the remaining pyramids are rooted stars, which are not caterpillar functions. We do not claim that their coefficients agree with ours. We establish explicit width bounds for the first and second hidden layers; their paper states no widths.

3 The Caterpillar Net

Two identities drive the construction:

$$\max(s, t) = t + \sigma(s - t), \tag{7}$$

$$\sigma(\ell(x)) = \sigma(-\ell(x)) + \ell(x). \tag{8}$$

The first suggests a path with attached leaves. A signed sum of the resulting caterpillar support functions agrees with the maximum modulo lower-dimensional terms; proper faces and linear functions correct the error. The second identity reverses hinges to merge units in Section 6.

Fix $N \geq 3$ and write $x = (x_0, \ldots, x_{N-1}) \in \mathbb{R}^N$.

Definition 3 (Labelled caterpillar, `Lean_3`)**.** A *caterpillar description* is a permutation–set pair

$$\pi \in \mathrm{Sym}\{0, \ldots, N-1\}, \qquad \varnothing \neq S \subseteq \{1, \ldots, r\}, \qquad r := N - 2. \tag{9}$$

Position j carries label $\pi(j+1)$, and $\pi(0)$ is the *apex*. The *spine positions* are $\{0\} \cup S$. The *cut set* S separates positions $j-1$ and j whenever $j \in S$. The last spine position q and the *parent* $p(j)$, the nearest preceding spine position, are

$$q := \max S, \qquad p(j) := \max\big(\{0\} \cup \{i \in S : i < j\}\big) \qquad (1 \leq j \leq r). \tag{10}$$

The parent–child edges form the *labelled caterpillar*: a path through the spine positions with each remaining label attached as a leaf to its parent (Figure 1). The apex is separate from the tree.

Definition 4 (Caterpillar function and face functions, `Lean_4`, `Lean_5`)**.** For labels i, j, for the tree edges, and for the head, put

$$\ell_{ij}(x) := x_i - x_j, \tag{11}$$

$$L_{\pi,S} := \big\{\ell_{\pi(p(j)+1),\pi(j+1)} : 1 \leq j \leq r\big\}, \tag{12}$$

$$h_{\pi,S} := \ell_{\pi(q+1),\pi(0)}. \tag{13}$$

The *edge functions* $L_{\pi,S}$ subtract each child's coordinate from its parent's, one for each of the r tree edges, and the *head function* $h_{\pi,S}$ compares the last spine vertex with the apex. For a linear function h, a finite set L of linear functions, and a *state* $\tau(\ell) \in \{0,\ \ell,\ \sigma \circ \ell\}$ chosen for every $\ell \in L$, define

$$U^{\tau}_{h,L}(x) := \sigma\Big(h(x) + \sum_{\ell \in L} \tau(\ell)(x)\Big). \tag{14}$$

Each edge is omitted, kept linearly, or kept as a hinge. Write $U_{h,L}$ for the all-hinge state, and call (h, L) its *function description*, recording the head and inner hinges. The *caterpillar function* of (π, S) and its *face functions* are

$$U_{\pi,S} := U_{h_{\pi,S},L_{\pi,S}}, \qquad U^{\tau}_{\pi,S} := U^{\tau}_{h_{\pi,S},L_{\pi,S}}. \tag{15}$$

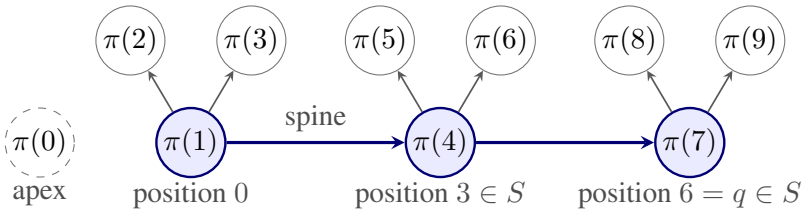


Figure 1: A labelled caterpillar with $N = 10$ and $S = \{3, 6\}$. The spine $\pi(1) \to \pi(4) \to \pi(7)$ has two leaves at each vertex; all arrows point from parent to child. The apex $\pi(0)$ is separate from the tree. The last spine position is $q = 6$, so the head function is $h_{\pi,S}(x) = x_{\pi(7)} - x_{\pi(0)}$.

The term *face function* comes from the following geometry. Identify $a(x) = \langle a, x\rangle$ with its coefficient vector. Then $\ell_{ij} = e_i - e_j$ points from e_j to e_i, and $h = h_{\pi,S} = e_{\pi(q+1)} - e_{\pi(0)}$ points from the apex label to the last spine label. Each $\ell \in L = L_{\pi,S}$ points from child to parent. As free vectors, h places a base corner and the ℓ generate its edges. The polytope apex is 0, distinct from the labelled point $e_{\pi(0)}$. Define the *support function* and the caterpillar polytope by

$$h_P(x) := \max_{p \in P} \langle p, x\rangle, \tag{16}$$

$$Q_{h,L} := \operatorname{conv}\Big(\{0\} \cup \Big(h + \sum_{\ell \in L} [0, \ell]\Big)\Big), \qquad U_{h,L} = h_{Q_{h,L}}. \tag{17}$$

Indeed, the segment $[0, \ell]$ has support function $\sigma \circ \ell$, Minkowski sums add support functions, and adjoining 0 applies the outer ReLU. The base is the translated zonotope $h + \sum_{\ell \in L}[0, \ell]$. For a state τ in Equation (14), set $t_\ell = \mathbb{1}_{\{\tau(\ell) = \ell\}}$, where $\mathbb{1}$ denotes the indicator. It is zero for omitted edges and hinges; the latter contribute segments. The corresponding polytope is $Q^{\tau}_{h,L} := \operatorname{conv}\big(\{0\} \cup (h + \sum_{\ell \in L} t_\ell \ell + \sum_{\tau(\ell) = \sigma \circ \ell}[0, \ell])\big)$, and $U^{\tau}_{h,L} = h_{Q^{\tau}_{h,L}}$. Figure 3 illustrates the translated generators for $N = 4$ on the slice $x_0 = 0$, where a becomes (a_1, a_2, a_3) and e_0 is represented by 0.

The head connects the apex label to the tree, so the head and $N-2$ edge functions are linearly independent. Thus the base is an $(N-2)$-dimensional zonotope with affine hull avoiding 0, and $Q_{h,L}$ is an $(N-1)$-dimensional pyramid. Each nontrivial face containing 0 cones over a base face. Maximizing on each base segment selects 0, ℓ, or $[0, \ell]$, with support functions 0, ℓ, and $\sigma \circ \ell$; the outer ReLU adjoins the apex. These are precisely the three states in Equation (14).

Definition 5 (Dictionary, `Lean_6`)**.** The *dictionary* $\mathcal{D}$ is the finite set of functions $\mathbb{R}^N \to \mathbb{R}$ consisting of every face function $U^\tau_{\pi,S}$, over all caterpillar descriptions (π, S) and all states τ, together with the *auxiliary functions*

$$\begin{aligned} &x \mapsto \sigma(x_i), \quad x \mapsto \sigma(-x_i) \quad (0 \le i < N), \\ &x \mapsto \sigma(x_i - x_j) \qquad\qquad\quad (0 \le i < j < N). \end{aligned} \tag{18}$$

Heads and edge functions are coordinate differences, so face functions and the functions $x \mapsto \sigma(x_i - x_j)$ are translation invariant: with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^N$,

$$U^\tau_{\pi,S}(x + t\mathbf{1}) = U^\tau_{\pi,S}(x) \qquad (t \in \mathbb{R}). \tag{19}$$

Only $\sigma(\pm x_i)$ change under this translation. Theorem 13 proves that $\mathcal{D}$ spans the maximum.

3.1 Rational coefficients

Let $V_N = \operatorname{span} \mathcal{D}$ and let A_N be the span of the auxiliary functions. Choose a basis $g_1, \dots, g_a$ of A_N from those functions and extend it by caterpillar face functions to a basis $g_1, \dots, g_d$ of V_N. Put

$$q_N = \dim(V_N / A_N) = d - a. \tag{20}$$

Choose rational inputs $y_1, \dots, y_d$ for which the evaluation matrix $E = (g_j(y_i))_{i,j}$ is invertible, and define

$$c = E^{-1}\big(\max_N(y_1), \dots, \max_N(y_d)\big)^\top. \tag{21}$$

These are the coefficients used by the Caterpillar Net.

Lemma 6 (The coefficient identity)**.** *The rational inputs can be chosen as stated, $c \in \mathbb{Q}^d$, and*

$$\max_N(x) = \sum_{j=1}^{d} c_j g_j(x) \qquad (x \in \mathbb{R}^N). \tag{22}$$

Proof. The vectors $(g_1(y), \dots, g_d(y))$, $y \in \mathbb{Q}^N$, span $\mathbb{R}^d$: otherwise a nonzero linear functional annihilating them would give a nontrivial combination of the g_j vanishing on the dense set $\mathbb{Q}^N$, hence everywhere by continuity. This contradicts independence. Select d independent evaluation vectors as the rows of E. All entries of E and the right side of Equation (21) are rational, so c is rational. By Theorem 13, $\max_N \in V_N$; its unique basis coordinates solve this same invertible system and therefore equal c. □

The first a terms form the auxiliary remainder R; the other q_N terms give the retained face functions. We retain all these basis functions, including any with zero coefficient. Distinct basis functions cannot normalize to the same description, so $|K| = q_N$ below. The following normalization and network assembly use these fixed rational coefficients.

3.2 The network

The network computes an expansion of $\max_N$ in face functions and auxiliary functions. To compile face functions, first absorb every linear edge of a face state into the head:

$$U^\tau_{h,L} = U_{h',M}, \qquad h' := h + \sum_{\tau(\ell)=\ell} \ell, \qquad M := \{\ell \in L : \tau(\ell) = \sigma \circ \ell\}. \tag{23}$$

By Equation (8), reversing each remaining ℓ_{ij} with $i > j$ gives the same function with the *normalized description*

$$M^+ := \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in M\},$$
$$h^+ := h' + \sum_{\ell_{ij}\in M,\ i>j} \ell_{ij}. \tag{24}$$

Geometrically, $[0, \ell] = \ell + [0, -\ell]$: reversing a generator shifts the head by the old generator. Normalization chooses a consistent orientation and a corresponding base corner; it does not rescale vectors or change the polytope (Figure 3). Write $U_k := U_{h,M}$ for a normalized description $k = (h, M)$, and let $\mathcal{K}_0$ be the set of normalized descriptions of all face functions. Use A_N as defined in Section 3.1. The network has three parts (Figure 2).

- The *first hidden layer* computes the shared features

$$\Phi(x) := \big(\sigma(x_0),\ \sigma(-x_0),\ (\sigma(x_i - x_0))_{i>0},\ (\sigma(x_i - x_j))_{i<j}\big). \tag{25}$$

- The *second hidden layer* computes the retained units U_k, $k \in K$, for a set $K \subseteq \mathcal{K}_0$, and the two neurons $\sigma(R)$ and $\sigma(-R)$ for an auxiliary remainder $R \in A_N$.
- The *output layer* returns $\sum_{k\in K} d_k U_k + \sigma(R) - \sigma(-R)$.

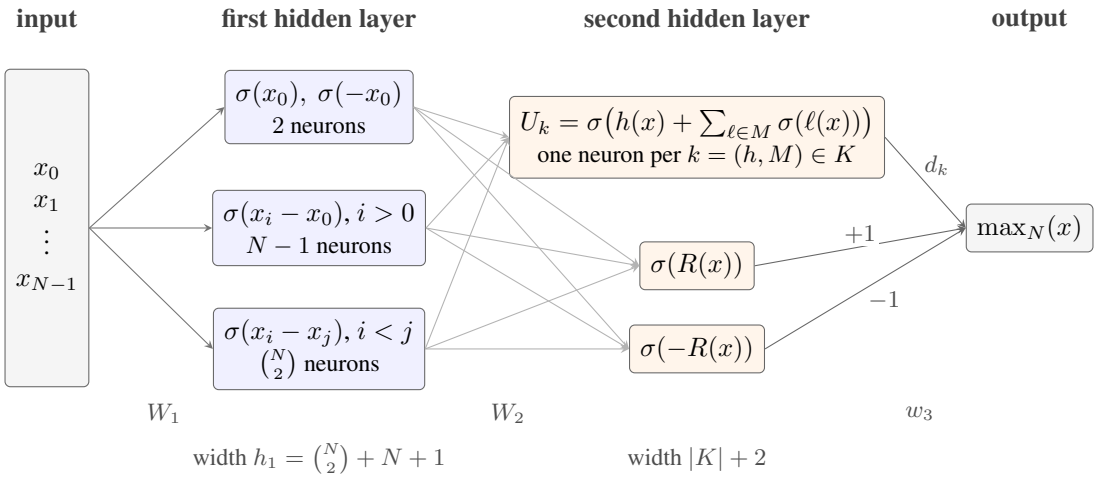


Figure 2: The Caterpillar Net. The first hidden layer is fixed. The second hidden layer has one neuron for every retained normalized description $k = (h, M)$ and two neurons for the auxiliary remainder R. All biases are zero; the matrices are given in Equation (32).

Lemma 7 (Network assembly, `Lean_7`, Proof)**.** *Normalize the q_N chosen face functions and collect their descriptions in $K \subseteq \mathcal{K}_0$. Let d_k be their coefficients from Equation* (21)*, and let R contain its auxiliary terms. Then*

$$\max_N = \sum_{k \in K} d_k U_k + R. \tag{26}$$

The network with first hidden layer Equation (25) *and the matrices in Equation* (32) *computes $\max_N$. It has widths h_1 and $|K| + 2 = q_N + 2$, rational weights, and zero biases.*

Proof of Lemma 7. The first hidden layer outputs Φ from Equation (25). Thus W_1 has rows $e_0^\top$ and $-e_0^\top$, then $e_i^\top - e_0^\top$ for $i > 0$, then $e_i^\top - e_j^\top$ for $i < j$, using a fixed ordering within each block. Its row count is

$$\binom{N}{2} + (N-1) + 2 = \binom{N}{2} + N + 1 = h_1. \tag{27}$$

By Equation (8), these features recover all coordinates,

$$\begin{aligned} x_0 &= \sigma(x_0) - \sigma(-x_0), \\ x_i &= x_0 + \sigma(x_i - x_0) - \sigma(x_0 - x_i) \qquad (i > 0). \end{aligned} \tag{28}$$

For each $k = (h, M) \in K$, substitute Equation (28) into $h(x) + \sum_{\ell \in M} \sigma(\ell(x))$ and collect the coefficients of $\Phi(x)$ into a row vector $w_k^\top$. Every hinge in M already has the orientation $i < j$, so its coefficient is added directly to the corresponding feature.

To specify the remaining row, collect the auxiliary coefficients of R as

$$R(x) = \sum_i p_i \sigma(x_i) + \sum_i n_i \sigma(-x_i) + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{29}$$

Each retained unit is translation invariant by Equation (19), and $\max_N(x + t\mathbf{1}) = \max_N(x) + t$, so Equation (26) gives

$$R(x + t\mathbf{1}) = R(x) + t, \qquad \sum_i p_i = 1. \tag{30}$$

The coefficient sum follows by setting $x = 0$ and $t = 1$. Now choose t large enough that every $x_i + t$ is nonnegative. Substituting the auxiliary expansion into Equation (30) cancels t and gives

$$R(x) = \sum_i p_i x_i + \sum_{i<j} s_{ij} \sigma(x_i - x_j). \tag{31}$$

Substitute Equation (28) into this expression and collect the coefficients of $\Phi(x)$ into w_R. Thus both w_k and w_R are specified by finite sums of the previously defined coefficients. The entries of W_1 and of the rows w_k are integers, because heads and hinges are integer combinations of coordinates. The row w_R is a rational combination of the p_i and s_{ij}, and w_3 consists of the d_k and ± 1.

Order K and set the second hidden layer matrix and output layer vector to

$$W_2 = \begin{pmatrix} (w_k^\top)_{k \in K} \\ w_R^\top \\ -w_R^\top \end{pmatrix}, \qquad w_3 = \begin{pmatrix} (d_k)_{k \in K} \\ 1 \\ -1 \end{pmatrix}, \qquad b_1 = b_2 = b_3 = 0. \tag{32}$$

The second hidden layer outputs $U_k(x)$ and $\sigma(\pm R(x))$. By Equation (8) and Equation (26),

$$w_3^\top \sigma(W_2 \Phi(x)) = \sum_{k \in K} d_k U_k(x) + \sigma(R(x)) - \sigma(-R(x)) = \max_N(x). \tag{33}$$

□

This completes the rational Caterpillar Net. Section 6 bounds q_N by counting its possible normalized descriptions.

4 Lower bounds

One local test proves both kinds of lower bound: at a tied maximum, take a directional germ and discard lower-dimensional support functions. For expansion size, every tie must be detected by some unit, so we count the ties each unit can detect. For dictionary dimension, we exhibit one locality that detects each path order separately.

Definition 8 (Caterpillar expansion, `Lean_8`)**.** A finite set $K \subseteq \mathcal{K}_0$ is a *caterpillar expansion* of $\max_N$ if $\max_N = \sum_{k\in K} d_k U_k + R$ for real coefficients d_k and $R \in A_N$. Its size is $|K|$.

Theorem 9 (Exponential count, Proof)**.** *Put $y_0 = \sqrt{3/2} - 1$. Every caterpillar expansion of $\max_N$, $N \geq 3$, satisfies*

$$|K| \geq (1+y_0)^N - 1 - Ny_0 - \binom{N}{2} y_0^2 = \Omega\big((3/2)^{N/2}\big). \tag{34}$$

The bound includes auxiliary remainders and arbitrary signed output coefficients. It also holds for broader tree dictionaries described below. It does not give a lower bound for unrestricted two-hidden-layer networks.

Theorem 10 (Vocabulary dimension, `Lean_9`, Proof)**.** *For every $N \geq 3$, $q_N = \dim(V_N/A_N) \geq (N-2)!$.*

Thus the whole dictionary has factorial dimension, and $\log q_N = \Theta(N \log N)$ by Equation (97). This concerns the basis used in Section 3.1; an individual expansion of the maximum may have smaller support.

Broader forest dictionaries. For a real head h and a forest of oriented edges M, write $U(x) = \sigma(\sum_i h_i x_i + \sum_{(i,j)\in M} \sigma(x_i - x_j))$ and $\kappa_i = h_i + |\{j : (i,j) \in M\}|$. Delete simultaneously every degree-one vertex with $\kappa_i = 0$; call the result the *reduced forest*.

Theorem 11 (Forest expansions, Proof)**.** *Suppose $\max_N = \sum_{k\in K} d_k U_k + R$, with $R \in A_N$, real heads, and reduced forests of maximum degree at most $\Delta \geq 2$. Set $y = (2\Delta+2)^{-1}$; when $\Delta = 2$, one may instead use $y = y_0 = \sqrt{3/2} - 1$. Then*

$$|K| \geq \sum_{m=3}^{N} \binom{N}{m} y^m = (1+y)^N - 1 - Ny - \binom{N}{2} y^2. \tag{35}$$

Caterpillar face functions have reduced degree at most two. The rooted stars $\sigma(x_c - x_a + \sum_{l\neq a,c} \sigma(x_c - x_l))$ and all their face functions have reduced degree zero. Consequently the stronger rate in Equation (34) counts every term in any mixture of our building blocks and those of the direct combinatorial construction in [BHR+26]; Appendix D identifies the latter exactly. The real-head extension and the stronger aggregate estimates are proved mathematically here; the existing Lean count theorem gives a weaker bound.

Arbitrary tree shapes with a source orientation. Let T be any tree on the labels other than an apex a, choose vertices s, r of T, and orient all edges away from s. The functions $U_{a,s,r,T}(x) = \sigma(x_r - x_a + \sum_{p\to q \text{ in } T} \sigma(x_p - x_q))$ may be adjoined to the reduced-degree-two family without changing Equation (34). Indeed, $\kappa_v = \deg_T(v) - 1 + \mathbf{1}_{v=s} + \mathbf{1}_{v=r}$, so only s and r can be degree-one vertices with $\kappa_v = 1$. The proof selects a degree-two covering subforest and uses the same block count. This includes every tree shape, with an arbitrary head root. It does not include arbitrary face states of these added trees: linearizing an edge can change the leaf coefficients.

Large matchings with arbitrary orientations and weights. Start with any mixture of the reduced-degree-two forest functions and the source-oriented functions above. Adjoin units

$$U_j(x) = \sigma\left(h_j(x) + \sum_{(i,l)\in M_j} c_{j,il}\sigma(x_i - x_l)\right), \tag{36}$$

where the heads are arbitrary real linear forms, the coefficients are arbitrary real numbers, and the comparison graph is a pseudoforest, meaning each connected component has at most one cycle. Suppose each added graph has a matching of size at least k, where k is a nonnegative integer with $2k \le N$. For $N \ge 9$, every expansion of the maximum in this mixture, with an arbitrary shallow remainder, uses a total number Z of terms satisfying

$$Z \ge \frac{(9/7)^k}{2d_k}, \qquad d_k = 1 + \frac{2k}{7} + \frac{4}{49}\binom{k}{2}. \tag{37}$$

Thus $k = \Omega(N)$ gives an exponential bound. Biases may also be allowed: taking the recession function preserves the homogeneous target and the comparison graphs.

Every orientation and face state of a canonical tree with matching number at least k is included: omitted edges can be retained with coefficient zero, and linear edge terms absorbed into the head. Thus the matching condition concerns the original tree. This includes perfect and near-perfect matchings and stars with every edge subdivided once. More generally, a tree on $n \ge 2$ vertices with L leaves has a matching of size at least $(n - L + 1)/2$, so a fixed positive fraction of nonleaf vertices suffices. Additional cycles with total excess $b = o(N)$ are also permitted when $k = \Omega(N)$; the proof gives the same exponential rate up to $\exp(o(N))$.

Corollary 12 (All tree orientations, `Lean_10`)**.** *For $N \ge 3$, every expansion of $\max_N$ by canonical tree face functions and a remainder in A_N has at least $N(N-1)/6$ terms. The head is $x_r - x_a$ for distinct root and apex labels; the tree on the other labels, all edge orientations, and all omitted, linear, or ReLU edge states are arbitrary. Output coefficients may have either sign.*

Adding all oriented stars. Canonical oriented star face functions can be added while counting only nonstar terms in the following exponential bounds. The base-family rate remains $\Omega((3/2)^{N/2})$, and Equation (37) holds for $N \ge 15$, counting only nonstar terms. This uses a fixed exact separator and coordinate deletion, described at the end of Appendix C.1. The unrestricted all-tree exponential bound remains open.

5 Why the dictionary spans the maximum

The maximum is the support function of a simplex. To work in its affine dimension, set

$$\Delta := \operatorname{conv}\{0, e_1, \dots, e_{N-1}\} \subseteq H_0 := \{x \in \mathbb{R}^N : x_0 = 0\}. \tag{38}$$

For every $x \in \mathbb{R}^N$,

$$\max_N(x) = x_0 + h_\Delta(x - x_0\mathbf{1}). \tag{39}$$

It suffices to prove that h_Δ lies in the span of the restrictions of $\mathcal{D}$ to H_0: the substitution $x \mapsto x - x_0\mathbf{1}$ fixes every face function, sends coordinate features to $\sigma(\pm(x_i - x_0))$, and preserves the dictionary span by Equation (8).

Henceforth all polytopes lie in H_0, and heads and edge functions are identified with their restricted coefficient vectors. Let $\mathcal{L}_k$ be the real span of support functions of polytopes of affine dimension at most k; thus $\mathcal{L}_0$ consists of linear functions. The *exposed face* of P in direction $u \in H_0$ is

$$P^u := \{p \in P : \langle p, u\rangle = h_P(u)\}. \tag{40}$$

The opposite face P^{-u} consists of the minimizers. Define

$$D_u h_P(x) := h_{P^u}(x) - h_{P^{-u}}(x) \qquad (x \in H_0). \tag{41}$$

Here u selects the faces, while x is the support-function input; $P^0 = P$ gives $D_0 = 0$. For a simplex facet normal, D_u subtracts the opposite vertex's linear support function from that of the facet.

For a finite linear combination g of support functions, define its one-sided directional derivative at u by

$$\begin{aligned} \partial_u g(x) &:= \lim_{t\downarrow 0} \frac{g(u+tx) - g(u)}{t}, \\ \partial_u h_P = h_{P^u}, &\qquad D_u g = \partial_u g - \partial_{-u} g. \end{aligned} \tag{42}$$

This makes D_u independent of the support-function decomposition. When $\langle u, \cdot\rangle$ is constant on P, the exposed faces cancel; otherwise both have smaller dimension. Consequently,

$$D_u \mathcal{L}_k \subseteq \mathcal{L}_{k-1} \qquad (k \geq 1). \tag{43}$$

The signed caterpillar sum. We seek a combination that agrees with h_Δ up to a function in $\mathcal{L}_{N-2}$. Adding $x_{\pi(0)}$ makes the caterpillar function the following maximum

$$x_{\pi(0)} + U_{\pi,S}(x) = \max\left\{ x_{\pi(0)},\, x_{\pi(q+1)} + \sum_{\ell \in L_{\pi,S}} \sigma(\ell(x)) \right\}. \tag{44}$$

Averaging over permutations gives the required symmetry

$$F_N(x) := \frac{1}{N!} \sum_{\pi} \sum_{\varnothing \neq S \subseteq \{1,\dots,r\}} (-1)^{N+|S|} \big(x_{\pi(0)} + U_{\pi,S}(x)\big), \tag{45}$$

The signs cancel the extra exposed-face contributions modulo $\mathcal{L}_{N-3}$, as proved below.

Theorem 13 (Dictionary sufficiency, `Lean_11`, Proof)**.** *For every $N \geq 3$, the dictionary of Definition 5 satisfies*

$$\max_N \in \operatorname{span} \mathcal{D}. \tag{46}$$

Proof of Theorem 13. Define the error by

$$f := F_N|_{H_0} - h_\Delta. \tag{47}$$

Step 13.1 reduces dictionary membership to $f \in \mathcal{L}_{N-2}$; Step 13.2 reduces this to $D_u f \in \mathcal{L}_{N-3}$ for all u. Step 13.3 proves these bounds by cancellation, using induction only when deleting a single exceptional coordinate.

Step 13.1 (Reduce to a lower-dimensional error). Assume temporarily that

$$f = F_N - h_\Delta \in \mathcal{L}_{N-2}. \tag{48}$$

Write $Q_{\pi,S} := Q_{h_{\pi,S},L_{\pi,S}}$. Removing the linear apex terms from Equation (45) gives

$$\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \in \mathcal{L}_{N-2}. \tag{49}$$

We correct the error by subtracting derivatives ∂_u until it is linear. Faces of $Q_{\pi,S}$ already have support functions in the dictionary span: apex–base faces give Equation (14), base faces give linear functions plus hinges, and the apex alone gives zero. Coordinate features and Equation (8) represent the linear terms and either hinge orientation. Differentiating h_Δ also produces simplex faces; induction on their vertex count supplies these functions.

Claim 13.1 [Simplex faces, `Lean_12`, `Lean_13`, Proof]. For every nonempty $T \subseteq \{0, e_1, \dots, e_{N-1}\}$, the support function of $\Delta_T := \operatorname{conv} T$ lies in the dictionary span.

Start ($|T| = 1$). The support function of a singleton is linear and lies in the dictionary span.

Step ($|T| \geq 2$). Fix T and assume the claim for all smaller nonempty vertex sets. For any current vertex set A with $|A| \geq 2$, remove $v \in A$ using a direction whose value at v exceeds its common value at the other vertices. Writing $\Delta_A := \operatorname{conv} A$, this gives

$$\begin{aligned} u_v &:= \begin{cases} -\sum_{i=1}^{N-1} e_i & v = 0, \\ v & v \neq 0, \end{cases} \\ (\Delta_A)^{u_v} &= \{v\}, \qquad (\Delta_A)^{-u_v} = \Delta_{A\setminus\{v\}}, \\ D_{u_v} h_{\Delta_A}(x) &= \langle v, x\rangle - h_{\Delta_{A\setminus\{v\}}}(x). \end{aligned} \tag{50}$$

Order the vertices outside T as $v_1, \dots, v_{N-|T|}$ and set

$$\varepsilon_T := D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} \left(\frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}h_{Q_{\pi,S}} - h_\Delta \right). \tag{51}$$

Each deletion changes the sign of the remaining simplex term, and subsequent D_u annihilate the linear terms. Hence

$$D_{u_{v_{N-|T|}}} \cdots D_{u_{v_1}} h_\Delta = (-1)^{N-|T|} h_{\Delta_T} + \ell_T, \tag{52}$$

where ℓ_T is linear. Thus the coefficient of h_{Δ_T} in ε_T is $(-1)^{N-|T|+1}$; all other terms are linear or caterpillar-face support functions and lie in $\operatorname{span}\mathcal{D}$. By Equation (43), $\varepsilon_T \in \mathcal{L}_{|T|-2}$. For the full vertex set the empty composition is the identity.

Lower the residual dimension recursively by

$$\begin{aligned} \varepsilon_T^{(0)}(x) &:= \varepsilon_T(x), \\ \varepsilon_T^{(j+1)}(x) &:= \varepsilon_T^{(j)}(x) - \sum_{u\in E_{T,j}} \big(\partial_u \varepsilon_T^{(j)}\big)(x). \end{aligned} \tag{53}$$

Here $0 \leq j < |T| - 2$; the next claim chooses each finite $E_{T,j} \subseteq H_0$.

Claim 13.2 [Residual reduction, `Lean_14`, Proof]. The sets $E_{T,j}$ can be chosen successively so that, for $0 \le j \le |T| - 2$,

$$\begin{aligned} \varepsilon_T^{(j)} &\in \mathcal{L}_{|T|-2-j}, \\ \varepsilon_T - \varepsilon_T^{(j)} &\in \operatorname{span} \mathcal{D}. \end{aligned} \tag{54}$$

Each residual remains a linear combination of h_{Δ_T}, support functions of its proper faces and of caterpillar faces, and linear functions.
The proof in Appendix A.1 chooses a direction orthogonal to each maximal-dimensional direction space, avoiding orthogonality to the simplex and other spaces. The derivative sum preserves each maximal term exactly once; subtracting it lowers the dimension.

At stage $j = |T| - 2$, the error is linear, so Equation (54) gives $\varepsilon_T \in \operatorname{span} \mathcal{D}$. Its simplex coefficient is ± 1, and its other terms already lie in the span; hence $h_{\Delta_T} \in \operatorname{span} \mathcal{D}$. Taking the full vertex set proves $h_\Delta \in \operatorname{span} \mathcal{D}$ under assumption Equation (48).

Step 13.2 (Why the facet bounds imply the class identity). For $N \ge 4$ and $f = F_N - h_\Delta$, we show that $f \in \mathcal{L}_{N-2}$ follows from

$$D_u f \in \mathcal{L}_{N-3} \qquad \text{for every } u \in H_0. \tag{55}$$

The base zonotope admits the Freudenthal–Kuhn triangulation [Fre42, Kuh60]; coning it to the apex expresses each caterpillar support function as a sum of simplex support functions modulo $\mathcal{L}_{N-2}$. For a simplex, D_u isolates a facet up to the linear support function of its opposite vertex. Thus Equation (55) controls the weighted sums of parallel facets. A common linear reconstruction operator turns these facet cancellations back into a cancellation of the original simplex sum. Define

$$(\mathcal{T} g)(x) := \begin{cases} g(x) & x_1 \ge 0, \\ g(x - x_1 e_1) & x_1 < 0. \end{cases} \tag{56}$$

Claim 13.3 [Simplex reconstruction, `Lean_15`, Proof]. Every $(N-1)$-dimensional simplex $P \subseteq H_0$ satisfies

$$h_P - \mathcal{T}\left(\sum_{F \text{ facet of } P} \operatorname{sgn}\langle \nu_F, e_1 \rangle \, h_F \right) \in \mathcal{L}_{N-2}. \tag{57}$$

Here ν_F is an outward normal of F. Its sign against e_1 records whether the facet bounds lines parallel to e_1 from above or below; parallel facets contribute zero. The operator $\mathcal{T}$ fills the segments between a polytope and its projection onto $p_1 = 0$, up to a segment correction for translation; hence $\mathcal{T}\mathcal{L}_k \subseteq \mathcal{L}_{k+1}$. Applying the claim with the same operator to every simplex preserves the cancellations between parallel facets. Their sums lie in $\mathcal{L}_{N-3}$, so their reconstructions lie in $\mathcal{L}_{N-2}$, as required. Appendix A.2 proves the claim and the subdivision and grouping steps.

Claim 13.4 [Class identity, `Lean_16`, Proof]. The class identity Equation (48) holds for every $N \ge 3$.

Step 13.3 (Prove the class identity by induction).

Start ($N = 3$). Here $S = \{1\}$, $q = 1$, and $p(1) = 0$, so Equation (44) gives

$$\begin{aligned} x_{\pi(0)} + U_{\pi,\{1\}}(x) &= \max\bigl\{ x_{\pi(0)}, x_{\pi(2)} + \sigma(x_{\pi(1)} - x_{\pi(2)}) \bigr\} \\ &= \max\{ x_{\pi(0)}, x_{\pi(1)}, x_{\pi(2)} \} = \max_3(x). \end{aligned} \tag{58}$$

All six terms have coefficient $1/6$, so

$$F_3 = \max_3 . \tag{59}$$

On H_0, this gives $f = 0 \in \mathcal{L}_1$.

Step ($N \geq 4$). Assume the class identity for $N-1$. By Step 13.2, it suffices to prove $D_u f \in \mathcal{L}_{N-3}$ for every $u \in H_0$. The linear apex terms disappear. The exposed simplex faces have the largest and smallest coordinate values u_i, including $u_0 = 0$. In a caterpillar base, $[0, \ell_{ij}]$ remains free when $u_i = u_j$; its maximizing endpoint is ℓ_{ij} when $u_i > u_j$ and 0 when $u_i < u_j$. Minimization reverses these choices.

Which faces can survive? Fix $u \in H_0 \setminus \{0\}$; the direction 0 gives $D_0 f = 0$. A base face is a zonotope, whose support function is linear plus a sum of segment support functions, hence belongs to $\mathcal{L}_1 \subseteq \mathcal{L}_{N-3}$. The apex alone contributes zero. If j tree edges have unequal endpoint values under u, an apex–base face has at most $N-2-j$ free segments and therefore dimension at most $N-1-j$. Thus only $j \leq 1$ needs attention.

Write $a = u_{\pi(0)}$. An apex–base face requires the maximum over the base to be zero. For $j = 0$, all tree values equal c, so this condition is $c - a = 0$, forcing $u = 0$. For $j = 1$, deleting the unequal edge leaves constant values $c \neq d$ on its two components. The zero-maximum condition is

$$a = \max\{c, d\} \qquad \text{for a spine edge,} \tag{60}$$

$$a = c + \sigma(c - d) \qquad \text{for a leaf edge, with leaf value } d. \tag{61}$$

The spine case gives two coordinate values. The leaf case does too when $d > c$; when $d < c$, it gives the values $c + t, c - t, c, \ldots, c$, where $t = c - d > 0$. Replacing u by $-u$ treats the opposite face. Consequently, only two-value directions and these symmetric three-value directions can contribute modulo $\mathcal{L}_{N-3}$.

Three values: reflection symmetry. Suppose $u_\alpha = c + t$, $u_\beta = c - t$, and all other coordinates equal c. Let τ exchange α and β. On H_0, its coordinate action is

$$(Rx)_i = x_{\tau(i)} - x_{\tau(0)}, \qquad Rx = x - \lambda(x)u, \qquad \lambda(x) := \frac{x_\alpha - x_\beta}{t}. \tag{62}$$

In particular $Ru = -u$. For $G = F_N|_{H_0}$, permutation symmetry and a common coordinate shift give $G \circ R = G + \ell$ with ℓ linear. Every exposed face in direction v lies in the hyperplane $\langle v, p \rangle = h_P(v)$; hence $\partial_v G(x + sv) = \partial_v G(x) + sG(v)$. Applying this to the derivative of $G \circ R = G + \ell$ at u gives

$$\partial_u G(x) + \ell(x) = \partial_{-u} G(Rx) = \partial_{-u} G(x) + \lambda(x) G(-u). \tag{63}$$

Therefore

$$D_u G = \lambda\, G(-u) - \ell \in \mathcal{L}_0. \tag{64}$$

The two exposed simplex faces are vertices, so $D_u f \in \mathcal{L}_0$ as well.

Handling two distinct values. First suppose that each value occurs at least twice. After the preceding dimension bound, every term still requiring cancellation has a unique split $u_{\pi(1)} = \cdots = u_{\pi(k)} \neq u_{\pi(k+1)} = \cdots = u_{\pi(r+1)}$ with $k \in S$. If the apex label has the latter value, toggle membership of $k-1$ in S. Otherwise, exchange the equal-valued labels $\pi(k+1), \pi(k+2)$ and toggle membership of $k+1$. Repetition of both values gives $k \geq 2$ in the first case and $k + 2 \leq r + 1$ in the second, so the operations are defined; the unchanged cut

at k keeps the cut set nonempty. The transformed term remains in the same case, and repeating the operation returns the original term. If Φ denotes this involution on the active terms and $w_{\pi,S} := (-1)^{N+|S|}/N!$, then

$$w_{\Phi(\pi,S)} = -w_{\pi,S}, \qquad D_u h_{Q_{\pi,S}} - D_u h_{Q_{\Phi(\pi,S)}} \in \mathcal{L}_1. \tag{65}$$

Inactive fixed points of the global transformation are already in $\mathcal{L}_{N-3}$ by the dimension bound. Summing the active pairs therefore puts $D_u F_N$ in $\mathcal{L}_{N-3}$. Each exposed simplex face has at most $N-2$ vertices and hence dimension at most $N-3$, so $D_u f \in \mathcal{L}_{N-3}$ without using induction.

One exceptional coordinate: the induction hypothesis. Suppose one of the two values occurs only once. Choose τ with $\tau(0)$ the exceptional label. To stay in H_0, relabel and subtract the new zeroth coordinate:

$$B_\tau : H_0 \longrightarrow H_0, \qquad (B_\tau x)_i := x_{\tau(i)} - x_{\tau(0)} \qquad (0 \le i < N). \tag{66}$$

Its inverse is $B_{\tau^{-1}}$, and it sends u to $(0, t, \dots, t)$, $t \neq 0$. Transforming both input and direction preserves every $\mathcal{L}_k$ and changes f only by a linear term, annihilated by D. Rescaling, and reversing the sign of D_u if needed, reduces to $u = (0, 1, \dots, 1)$.

This direction exposes the $(N-2)$-dimensional facet $\Delta' := \mathrm{conv}\{e_1, \dots, e_{N-1}\}$ and the opposite vertex 0. Hence

$$D_u h_\Delta(x) = h_{\Delta'}(x) = \max\{x_1, \dots, x_{N-1}\} \qquad (x \in H_0). \tag{67}$$

The induction hypothesis also applies to unrestricted lower-arity inputs modulo linear functions. Indeed, put $z = (x_1, \dots, x_{N-1})$ and $y = z - x_1 \mathbf{1}$. Each summand of F_{N-1} increases by the common shift, so

$$F_{N-1}(z) - \max_{N-1}(z) \equiv F_{N-1}(y) - \max_{N-1}(y) \pmod{\mathcal{L}_0}. \tag{68}$$

The first coordinate of y is zero. Induction and linear precomposition therefore give

$$\big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big) \in \mathcal{L}_{N-3}. \tag{69}$$

Claim 13.5 [Coordinate deletion, `Lean_17`, Proof]. For $N \ge 4$,

$$D_{(0,1,\dots,1)}(F_N|_{H_0}) - \big(x \mapsto F_{N-1}(x_1, \dots, x_{N-1})\big) \in \mathcal{L}_{N-3}. \tag{70}$$

The proof is in Appendix A.3. Its mechanism is that only degree-one tree positions of label 0 survive. Each nonempty lower cut set has N insertion preimages, each contributing $1/N$ of its lower coefficient. The extra empty-cut star sum lies in $\mathcal{L}_{N-3}$ by Boolean cancellation.

Completion of the induction. Combining Equation (67), Equation (69), and Claim 13.5 gives

$$\begin{aligned} D_u f(x) = {} & \big[D_u(F_N|_{H_0})(x) - F_{N-1}(x_1, \dots, x_{N-1})\big] \\ & + \big[F_{N-1}(x_1, \dots, x_{N-1}) - \max\{x_1, \dots, x_{N-1}\}\big]. \end{aligned} \tag{71}$$

Both brackets lie in $\mathcal{L}_{N-3}$, the first by Claim 13.5 and the second by induction. Step 13.2 now gives $f \in \mathcal{L}_{N-2}$, and Step 13.1 gives the dictionary representation. □

6 Network size

The widths in Theorem 1 are worst-case upper bounds: we count all possible descriptions after collecting linear terms into the head and orienting the remaining hinges consistently. The construction retains q_N basis functions; zero padding realizes the stated widths.

The normalized description Equation (24) of a caterpillar function depends only on its undirected spine, apex, and leaf attachments. For m spine vertices these choices give the summand of C_N in Equation (6). Each of the $N-2$ generators has three face states, giving at most $3^{N-2}C_N$ descriptions. The $N(N-1)$ descriptions with no retained hinge and head $x_i - x_j$, $i \neq j$, can be combined with the auxiliary functions. By Lemma 7, the second width is the number of retained descriptions plus two; Appendix B gives the complete count.

Formalization (`Lean_18`). In this working draft, the theorem and equation links point to the local Lean sources. Appendix E prints the network definitions and the main theorem, and describes the verification certificate.

7 CPWL functions

By the Wang–Sun theorem, every CPWL function on $\mathbb{R}^d$ is a signed sum of maxima of $d+1$ affine functions. These affine functions can be taken from any finite family among which the function selects pointwise, without rescaling them.

Theorem (Koutschan–Moser–Ponomarchuk–Schicho, [KMPS23, Theorem 2.4], `Lean_19`, `Lean_20`)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be continuous, and let $g_1, \dots, g_n : \mathbb{R}^d \to \mathbb{R}$ be affine with $f(x) \in \{g_1(x), \dots, g_n(x)\}$ for every x. Then there are $T \in \mathbb{N}$, signs $\varepsilon_t \in \{-1, 1\}$, and maps $\alpha_t : \{0, \dots, d\} \to \{1, \dots, n\}$ with*

$$f(x) = \sum_{t=1}^{T} \varepsilon_t \max_{d+1}\big(g_{\alpha_t(0)}(x), \dots, g_{\alpha_t(d)}(x)\big) \qquad \text{for every } x \in \mathbb{R}^d. \tag{72}$$

The affine functions of every polyhedral description of a CPWL function satisfy the hypothesis. The Lean proof follows Ovchinnikov's max–min form $f = \max_S \min_{j \in S} g_j$, expands it by inclusion–exclusion into an integer combination of maxima over subsets of $\{g_1, \dots, g_n\}$, and reduces each maximum to subsets of at most $d+1$ functions by an affine dependence among $d+2$ gradients.

Reusing the first hidden layer (`Lean_21`, `Lean_22`). Let $d \geq 2$, and let (h_1, h_2) be the widths of Theorem 1 for $N = d+1$. Substituting the arguments of each term of Equation (72) into the Caterpillar Net computes f. All terms use only the features $\sigma(g_a - g_b)$ for $a < b$, $\sigma(\pm g_1)$, and $\sigma(g_a - g_1)$, so one first hidden layer of width $\binom{n}{2} + n + 1$ serves every term. Terms with the same argument set $\alpha_t(\{0, \dots, d\})$ compute the same maximum and are merged, and the auxiliary remainders of all terms form one pair $\sigma(\pm R)$. The second hidden layer has width $s(h_2 - 2) + 2$, where s is the number of distinct argument sets and $s \leq \min\big(T, \sum_{m \leq d+1} \binom{n}{m}\big)$. Thus the second hidden layer holds one copy of the Caterpillar Net units for every distinct maximum; only the first hidden layer and the remainder pair are genuinely shared. The second hidden layer and output layer have rational weights and zero biases, and every first hidden layer neuron applies σ to $\pm(g_a - g_b)$ or $\pm g_a$. Hence rational g_i give rational weights and biases, and linear g_i give zero biases.

8 Conclusion

The Caterpillar Net uses q_N caterpillar face functions with rational coefficients, giving second hidden layer width $q_N + 2$ before padding. Conversely, every such expansion needs $\Omega((3/2)^{N/2})$ caterpillar functions (Theorem 9), so caterpillar building blocks cannot give subexponential second widths. Reducing our upper bound $2^{O(N \log N)}$ to $2^{O(N)}$ remains open. The factorial dimension bound in Theorem 10 limits compression by a basis of the entire dictionary; it does not give a factorial lower bound for representing the single function $\max_N$. Whether the maximum has two-hidden-layer representations of subexponential width with other second-layer functions remains open.

A Supporting proofs for the dictionary theorem

A.1 Residual reduction

We prove the inner induction used in the simplex-face argument of Section 5. Fix a nonempty vertex set T with $|T| \geq 2$, and assume that every proper simplex face on a subset of T is already in the dictionary span, as in the outer induction there.

Proof of Claim 13.2. We argue by induction on j.

Start ($j = 0$). By definition, $\varepsilon_T^{(0)} = \varepsilon_T$. Its dimension bound and support-function expression were established above, and the removed part is zero.

Step ($j < |T| - 2$). Assume the directions for the first j stages have been chosen and the claim holds at stage j. Put $k := |T| - 2 - j \geq 1$. We require each new direction u to be nonorthogonal to the simplex direction space $\operatorname{dir} \Delta_T := \operatorname{span}\{v - w : v, w \in T\}$. Its values on T are then not all equal, so the maximizing vertices form a nonempty proper subset:

$$\begin{aligned} A_u &:= \{v \in T : \langle u, v\rangle = h_{\Delta_T}(u)\}, \qquad \varnothing \neq A_u \subsetneq T, \\ \partial_u h_{\Delta_T} &= h_{\Delta_{A_u}} \in \operatorname{span} \mathcal{D}. \end{aligned} \tag{73}$$

The last membership uses the outer induction hypothesis, since $|A_u| < |T|$. An orthogonal direction would leave h_{Δ_T} unchanged and would not allow this use of the hypothesis.

Choose an auxiliary expression for $\varepsilon_T^{(j)}$ by support functions of polytopes of dimension at most k, and collect their distinct k-dimensional direction spaces. For each such space V, choose $u_V \in V^\perp$ not orthogonal to $\operatorname{dir} \Delta_T$ or to any other collected space V'. This is possible because $\dim \Delta_T = |T| - 1 > k$: each of $\operatorname{dir} \Delta_T$ and V' contains a vector outside V, whose nonzero projection onto $V^\perp$ is not orthogonal to that vector. Adding sufficiently small multiples of these projections avoids each orthogonality condition in turn while preserving those already avoided. Choose $E_{T,j} := \{u_V : V \text{ is a collected direction space}\}$, with $E_{T,j} = \varnothing$ if there are none. In the sum of derivatives over $E_{T,j}$, each term coming from a k-dimensional polytope is preserved exactly once. Every other derivative of that term is the support function of a proper face, while terms of lower dimension remain in $\mathcal{L}_{k-1}$. Subtracting this derivative sum therefore cancels all k-dimensional terms. By the recursive definition Equation (53), the resulting error satisfies $\varepsilon_T^{(j+1)} \in \mathcal{L}_{k-1} = \mathcal{L}_{|T|-3-j}$.

Now compute these same derivatives using the support-function expression in the claim. The derivative of h_{Δ_T} lies in the dictionary span by Equation (73). Differentiating the support function of an already proper simplex face still gives a support function of a proper face of Δ_T, so the same induction hypothesis applies. Differentiating the support function of a face of $Q_{\pi,S}$ gives a support function of a face of the same polytope, whose support functions lie in the dictionary span, and linear functions remain linear. Thus the derivative sum lies in the dictionary span. By Equation (53), $\varepsilon_T - \varepsilon_T^{(j+1)}$ is the sum of the previously removed part and this derivative sum, so it also lies in the dictionary span. Subtracting the derivative sum preserves the stated types of support-function terms. This completes the proof step.

□

A.2 Reconstruction from opposite exposed faces

We justify Step 13.2: for $N \geq 4$, the bounds Equation (55) imply $f \in \mathcal{L}_{N-2}$. We first subdivide the caterpillar polytopes, then prove Claim 13.3, and finally apply it to the resulting weighted simplex sum.

The base of each $Q_{\pi,S}$ is an affine cube. Order its edge vectors by a bijection $\omega : \{1, \dots, N-2\} \to L_{\pi,S}$, add them one at a time, and adjoin the apex:

$$K_{\pi,S,\omega} := \operatorname{conv}\left(\{0\} \cup \left\{h_{\pi,S} + \sum_{a=1}^{k} \omega(a) : 0 \le k \le N-2\right\}\right). \tag{74}$$

These $(N-2)!$ simplices are obtained by coning the *Freudenthal–Kuhn triangulation* of the base to the apex [Fre42, Kuh60]: in cube coordinates, each base simplex occupies one ordering region $1 \ge t_{\omega(1)} \ge \cdots \ge t_{\omega(N-2)} \ge 0$. For a hyperplane cut $P = P_+ \cup P_-$, the support-function identity

$$h_P + h_{P_+ \cap P_-} = h_{P_+} + h_{P_-} \tag{75}$$

shows that subdivision changes the sum of support functions only by support functions of the cutting sections. Hence

$$h_{Q_{\pi,S}} - \sum_{\omega} h_{K_{\pi,S,\omega}} \in \mathcal{L}_{N-2}. \tag{76}$$

Substituting Equation (76) into Equation (45) expresses f as a finite linear combination of support functions of $(N-1)$-dimensional simplices in H_0, plus a remainder in $\mathcal{L}_{N-2}$. The simplex Δ contributes its support function with coefficient -1, and the linear apex terms are included in the remainder.

For a simplex P, an outward facet normal ν_F exposes the facet F on one side and its opposite vertex v_F on the other. Thus

$$D_{\nu_F} h_P(x) = h_F(x) - \langle v_F, x \rangle. \tag{77}$$

Directions exposing no facet on either side give support functions of faces of dimension at most $N-3$. Consequently, Equation (55) says that each weighted, signed sum of parallel-facet support functions belongs to $\mathcal{L}_{N-3}$.

Our assumption controls the weighted facet contributions, whereas our conclusion concerns f. To recover the original sum modulo $\mathcal{L}_{N-2}$, we use the same linear operator for every simplex, so cancellations between facet contributions survive reconstruction. Use the operator $\mathcal{T}$ defined in Equation (56). The sign in this definition concerns the input x. To interpret it geometrically, first suppose that every point $p \in P$ satisfies $p_1 \ge 0$, and write $\operatorname{pr}(p) := p - p_1 e_1$ for projection onto $p_1 = 0$. Since $\langle p, x \rangle - \langle \operatorname{pr}(p), x \rangle = p_1 x_1$, each point dominates its projection when $x_1 \ge 0$, and its projection dominates when $x_1 < 0$. Consequently,

$$\mathcal{T} h_P(x) = \max\{h_P(x), h_{\operatorname{pr} P}(x)\} = h_{\operatorname{conv}(P \cup \operatorname{pr} P)}(x). \tag{78}$$

This convex hull fills the segments with endpoints p and its projection, for $p \in P$, so its dimension is at most $\dim P + 1$. For an arbitrary P, choose $c \ge 0$ such that $P + ce_1$ lies in $p_1 \ge 0$. Translation adds cx_1 to the support function, and $\mathcal{T}(x_1) = \sigma(x_1)$, giving

$$\mathcal{T} h_P(x) = h_{\operatorname{conv}((P+ce_1) \cup \operatorname{pr}(P+ce_1))}(x) - c\sigma(x_1). \tag{79}$$

The correction $\sigma(x_1)$ is the support function of the segment $[0, e_1]$. Thus linearity gives

$$\mathcal{T} \mathcal{L}_k \subseteq \mathcal{L}_{k+1}. \tag{80}$$

Proof of Claim 13.3. Let $k = |\{F \text{ facet of } P : \langle \nu_F, e_1 \rangle \ne 0\}|$ count facets not parallel to e_1. Boundedness ensures both nonzero signs occur, so $k \ge 2$.

Start ($k = 2$). There is one upper facet F^+ and one lower facet F^-. Every line parallel to e_1 that meets P has its two intersection endpoints on these facets. Thus $h_P(x) = h_{F^+}(x)$ when $x_1 \geq 0$, and $h_P(x) = h_{F^-}(x)$ when $x_1 < 0$. When $x_1 \geq 0$, $\mathcal{T}$ evaluates at x, so subtracting $\mathcal{T}(h_{F^+} - h_{F^-})$ from h_P leaves h_{F^-}. When $x_1 < 0$, $\mathcal{T}$ evaluates both facet support functions at $x - x_1 e_1$, where they agree because the facets have the same projection. Their difference is therefore zero, again leaving h_{F^-}. The reconstruction error is therefore $h_{F^-} \in \mathcal{L}_{N-2}$, proving the claim for $k = 2$.

Step ($k > 2$). Assume the claim for $k - 1$, and let P have k facets not parallel to e_1. Two of them have the same nonzero sign of $\langle \nu_F, e_1 \rangle$. Their intersection is an $(N-3)$-dimensional face C. Let $W := \operatorname{span}\{p - q : p, q \in C\}$, and project orthogonally within H_0 onto $W^\perp$. This is a two-dimensional plane, since $\dim H_0 = N - 1$ and $\dim W = N - 3$. All points of C have the same projection c, because their differences lie in W. There are just two vertices of P outside C, so the projected simplex is a triangle with vertices c and the projections of those two vertices. The chosen facets project to its two sides meeting at c.

The chosen facet normals are orthogonal to W, so their signs against e_1 equal their signs against its projection. This projected direction is nonzero, since the chosen facets are not parallel to e_1. If both sides are upper, moving from c opposite to this direction enters the triangle; if both are lower, moving in this direction enters it. Continuing meets the interior of the opposite edge. The inverse image of the line through c in this direction is a hyperplane in H_0 containing C and parallel to e_1. It cuts the edge whose endpoints are the two vertices outside C at an interior point and divides P into two $(N-1)$-dimensional simplices. Each keeps one of the two chosen facets and replaces the other by the new facet parallel to e_1. Its other facets retain their normal directions, so each new simplex has exactly $k - 1$ facets not parallel to e_1.

Write $E(Q)$ for the reconstruction error on the left of Equation (57). Let P_+ and P_- be the two new simplices, let $R := P_+ \cap P_-$ be their common cutting facet, and write F_1, F_2 for the chosen facets and $R_F := R \cap F$ for every other facet of P. With $s_F := \operatorname{sgn}\langle \nu_F, e_1 \rangle$, applying Equation (75) to P and to every split facet gives the exact residual identity

$$E(P) = E(P_+) + E(P_-) - h_R + \mathcal{T}\left(\sum_{\substack{F \text{ facet of } P \\ F \notin \{F_1, F_2\}}} s_F h_{R_F} \right). \tag{81}$$

Indeed, the inherited facets retain their signs, while the two copies of R are parallel to e_1 and therefore have sign zero. By the induction hypothesis, $E(P_+), E(P_-) \in \mathcal{L}_{N-2}$; moreover, $\dim R = N - 2$ and $\dim R_F \leq N - 3$. Equation Equation (80) therefore puts every term on the right of Equation (81) in $\mathcal{L}_{N-2}$. This proves the claim for k and completes the induction.

□

Conclusion for f. Apply Equation (57) to every simplex support-function term in the expansion of f, keeping its original coefficient. A facet F parallel to e_1 has $\langle \nu_F, e_1 \rangle = 0$, so its term in the reconstruction formula is zero and can be omitted. For each remaining occurrence of a facet, choose ν_F outward from its simplex and put

$$u_F := \frac{\nu_F}{\langle \nu_F, e_1 \rangle}, \qquad \langle u_F, e_1 \rangle = 1. \tag{82}$$

Thus the coordinate with index 1 is normalized to 1; the coordinate with index 0 remains zero because $u_F \in H_0$. Oppositely oriented parallel normals give the same u_F. Only the normal is rescaled: the support function and its coefficient are unchanged.

Let $\mathcal{U}$ be the finite set of vectors u_F obtained this way. For each $u \in \mathcal{U}$, define G_u by adding all facet support-function terms with $u_F = u$, retaining their signs and weights. An occurrence of F as a

facet of $K_{\pi,S,\omega}$ contributes $(-1)^{N+|S|}\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F/N!$, while an occurrence as a facet of Δ contributes $-\operatorname{sgn}\langle \nu_F, e_1\rangle\, h_F$. These are precisely the terms with the same normalized normal, including all repetitions.

More explicitly, define ρ to be the sum of the linear apex terms and the weighted subdivision errors in Equation (76). Then

$$f = \frac{1}{N!}\sum_{\pi}\sum_{\varnothing\neq S\subseteq\{1,\dots,N-2\}}(-1)^{N+|S|}\sum_{\omega} h_{K_{\pi,S,\omega}} - h_\Delta + \rho, \qquad \rho\in\mathcal{L}_{N-2}. \tag{83}$$

For a simplex P, collect its facets with normalized normal u as

$$B_u(P) := \sum_{\substack{F \text{ facet of } P\\ \langle \nu_F, e_1\rangle\neq 0,\ u_F = u}} \operatorname{sgn}\langle \nu_F, e_1\rangle h_F. \tag{84}$$

An empty sum is zero. If $\langle \nu_F, e_1\rangle > 0$, the direction u_F exposes F; if it is negative, $-u_F$ exposes F. The opposite face is a single vertex, so Equation (77) gives the signed facet term up to a linear function. If neither u nor $-u$ exposes a facet of P, both exposed faces have dimension at most $N-3$. Thus $D_u h_P - B_u(P)\in\mathcal{L}_{N-3}$. Applying this to the full weighted expansion, with the same permutations and nonempty cut sets as in Equation (45), gives

$$\begin{aligned} D_u f - G_u = &\sum_{\pi,S\neq\varnothing}\frac{(-1)^{N+|S|}}{N!}\sum_{\omega}\big(D_u h_{K_{\pi,S,\omega}} - B_u(K_{\pi,S,\omega})\big)\\ &- (D_u h_\Delta - B_u(\Delta)) + D_u\rho \in \mathcal{L}_{N-3} \qquad (u\in\mathcal{U}). \end{aligned} \tag{85}$$

Together with Equation (55), relation Equation (85) puts every G_u in $\mathcal{L}_{N-3}$. By linearity, the sum of the reconstructed facet terms is $\mathcal{T}(\sum_{u\in\mathcal{U}} G_u)$, which lies in $\mathcal{L}_{N-2}$ by Equation (80). Its difference from f consists of the reconstruction and subdivision remainders, also in $\mathcal{L}_{N-2}$. Hence $f\in\mathcal{L}_{N-2}$. This proves the implication used in Step 13.2.

A.3 Coordinate deletion

Proof of Claim 13.5. We now determine which terms of the signed sum remain for $u = (0,1,\dots,1)$. Only label 0 has a different coordinate value from the others. Consequently, a tree-edge generator is nonorthogonal to u exactly when one of its endpoint labels is 0.

If label 0 is the apex label, all tree-edge generators are orthogonal to u. The maximizing face is the whole base $B_{\pi,S}$ and the minimizing face is $\{0\}$, so $D_u h_{Q_{\pi,S}} = h_{B_{\pi,S}}\in\mathcal{L}_1\subseteq\mathcal{L}_{N-3}$. If label 0 is a tree vertex of degree at least two, at least two segment generators are nonorthogonal to u. The preceding dimension bound then gives $D_u h_{Q_{\pi,S}}\in\mathcal{L}_{N-3}$ as well. We can therefore discard both kinds of term: it suffices to retain descriptions in which label 0 is a tree vertex of degree one.

To express this condition in the permutation notation, let $p := \pi^{-1}(0)$ be the index of label 0 in the full order $(\pi(0),\dots,\pi(N-1))$. When $p\geq 1$, its tree position is $p-1$. A degree-one vertex is either an endpoint of the spine with no attached leaves or a leaf outside the spine. The retained descriptions are thus exactly

$$\begin{aligned} &p = 1, \quad 1\in S && \text{(root endpoint)},\\ &2\leq p\leq N-1, \quad p-1\notin S && \text{(leaf outside the spine)},\\ &p = N-1, \quad N-2\in S && \text{(last spine endpoint)}. \end{aligned} \tag{86}$$

Indeed, the root has only one child precisely when $1\in S$; a nonroot spine vertex has a parent and also a child unless it occupies the final tree position. Deleting label 0 from the full order now gives an order $\bar\pi$ mapping positions $\{0,\dots,N-2\}$ bijectively to labels $\{1,\dots,N-1\}$.

For $0 \le i \le N-3$, let ι_i insert a gap at the prospective pendant position $i+1$. Given a lower cut set $T \subseteq \{1,\dots,N-3\}$, the three active insertion rules are

$$\begin{aligned} \iota_i(t) &:= \begin{cases} t, & t < i+1, \\ t+1, & t \ge i+1, \end{cases} \\ \text{first end:} \quad & (p,S) = (1, \{1\} \cup \{t+1 : t \in T\}), \\ \text{pendant } i\text{:} \quad & (p,S) = (i+2, \{\iota_i(t) : t \in T\}) \qquad (0 \le i \le N-3), \\ \text{last end:} \quad & (p,S) = (N-1, T \cup \{N-2\}). \end{aligned} \tag{87}$$

In every row the upper permutation is uniquely obtained by inserting label 0 at position p in $\bar\pi$. Conversely, with t restricted to $\{1,\dots,N-3\}$, the deletion rules are $T = \{t : t+1 \in S\}$ for the first end, $T = \{t : \iota_i(t) \in S\}$ for pendant i, and $T = S \setminus \{N-2\}$ for the last end. These operations are inverse. The three cases in Equation (86) correspond respectively to the first-end row, the pendant row with $i = p-2$, and the last-end row. Thus a nonempty T has $N-2$ pendant and two endpoint preimages, whereas $T = \varnothing$ has only the two endpoints because upper cut sets are nonempty.

Extend the lower caterpillar notation to $T = \varnothing$ by setting its last spine position and every parent equal to 0, so its head begins at the root $\bar\pi(1)$. For every lower label order $\bar\pi$ put

$$\begin{aligned} C_{\bar\pi,T}(x) &:= x_{\bar\pi(0)} + U_{\bar\pi,T}(x), \\ q_T &:= \max(\{0\} \cup T), \qquad p_T(j) := \max(\{0\} \cup \{t \in T : t < j\}). \end{aligned} \tag{88}$$

For each upper preimage (π, S) of $(\bar\pi, T)$, the insertion identities and the literal factorial coefficients are

$$\begin{aligned} \text{pendant:} \quad & \frac{(-1)^{N+|S|}}{N!} = -\frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \quad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) + C_{\bar\pi,T} \in \mathcal{L}_1, \\ \text{endpoint:} \quad & \frac{(-1)^{N+|S|}}{N!} = \frac{1}{N}\frac{(-1)^{N-1+|T|}}{(N-1)!}, \qquad D_u\big(x_{\pi(0)} + U_{\pi,S}\big) - C_{\bar\pi,T} \in \mathcal{L}_1. \end{aligned} \tag{89}$$

Thus every one of the $N-2+2 = N$ preimages contributes $1/N$ of the lower summand modulo $\mathcal{L}_{N-3}$, and together they recover its coefficient in F_{N-1}.

The extra terms are the two endpoint fibres whose deletion leaves $T = \varnothing$, hence a rooted star. By Equation (89), each contributes $1/N$ of the weighted lower summand modulo $\mathcal{L}_1$. For a fixed lower order $\bar\pi$, the nonroot positions are $J = \{0, 2, \dots, N-2\}$, including the original apex position 0. For $a \in J$, let $\bar\pi^{(a)}$ exchange positions 0 and a, keeping the root at position 1; the exchange at $a = 0$ is the identity. Put

$$\begin{aligned} q := |J| = N-2, \qquad & d_a := x_{\bar\pi(1)} - x_{\bar\pi(a)} \quad (a \in J), \\ T_A := \sigma\left(\sum_{a\in A} d_a + \sum_{a \in J\setminus A} \sigma(d_a)\right) \quad & (A \subseteq J), \\ C_{\bar\pi^{(a)},\varnothing} = x_{\bar\pi(a)} + T_{\{a\}} \quad & (a \in J). \end{aligned} \tag{90}$$

The Boolean alternating identity gives

$$\sum_{A\subseteq J} (-1)^{|A|} T_A = 0 \quad \Longrightarrow \quad \sum_{a\in J} T_{\{a\}} = T_\varnothing + \sum_{\substack{A\subseteq J \\ |A|\ge 2}} (-1)^{|A|} T_A. \tag{91}$$

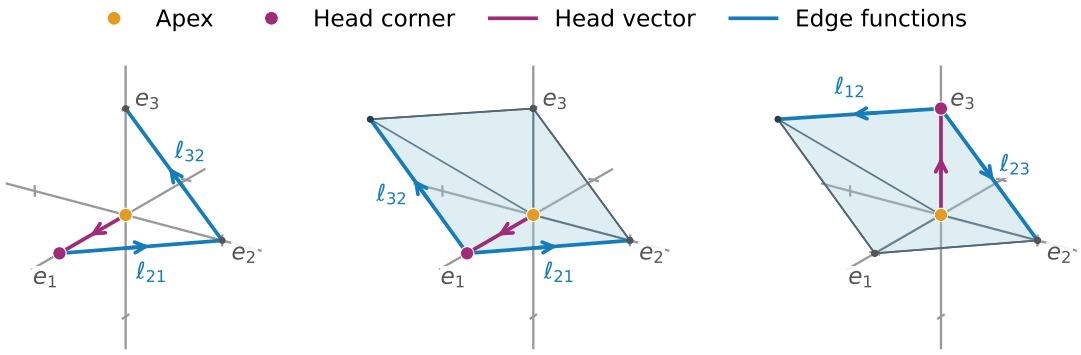
The empty term is a sum of support functions of segments. For $|A| \ge 2$, the polytope with support function T_A has one head direction and $q - |A|$ segment directions, hence dimension at most $1 + q - |A| \le q - 1 = N-3$.

Thus the singleton sum lies in $\mathcal{L}_{N-3}$; the sum of the linear apex coordinates lies in $\mathcal{L}_0$. Each fixed-position exchange is a bijection on lower label orders, so

$$q \sum_{\bar{\pi}} C_{\bar{\pi},\varnothing} = \sum_{\bar{\pi}} \sum_{a \in J} C_{\bar{\pi}^{(a)},\varnothing} \in \mathcal{L}_{q-1}. \tag{92}$$

Since $q > 0$, the full empty-cut sum belongs to $\mathcal{L}_{q-1} = \mathcal{L}_{N-3}$. The empty-cut coefficient $(-1)^{N-1}/(N-1)!$ and the factor $2/N$ from the two endpoint fibres are scalars, so the extra star sum lies there as well. This proves Claim 13.5. □

B Network construction and width bounds



(a) Coordinate differences (b) Original description (c) Normalized description

Figure 3: Coordinate differences and the caterpillar polytope for $\pi = (0, 3, 2, 1)$ and $S = \{1, 2\}$. (a) The head $h = e_1 - e_0$ and edge functions $L = \{\ell_{32} = e_3 - e_2, \ell_{21} = e_2 - e_1\}$ drawn between their labelled endpoints. (b) The same generators translated to the base corner h of $Q_{h,L}$. (c) Normalization (Section 6) gives $h^+ = h + \ell_{32} + \ell_{21} = e_3 - e_0$ and $L^+ = \{\ell_{12} = e_1 - e_2, \ell_{23} = e_2 - e_3\}$, leaving the polytope unchanged. Purple arrows represent the head (the apex–last spine connection before normalization); blue arrows represent the spine edge functions. Orange dots mark the polytope apex and purple dots the chosen head corner. The grey coordinate axes meet at the origin, representing e_0 after restriction to $x_0 = 0$; the positive unit points are labelled e_1, e_2, e_3.

Proof of Theorem 1. Fix $N \geq 3$. The construction in Sections 3.1 and 3.2 already gives rational weights and zero biases. We bound its second hidden layer width $q_N + 2$ by counting normalized descriptions.

Step 1.1 (Count the caterpillar face functions). Use the adjusted heads and normalized descriptions from Section 6. Counting distinct normalized descriptions bounds q_N.

For a caterpillar description (π, S), the spine endpoints are labelled $\pi(1)$ and $\pi(q+1)$, and every label other than $\pi(1)$ and the apex is the child of exactly one edge. Hence the normalized description of $U_{\pi,S}$ is

$$\begin{aligned} h^+_{\pi,S}(x) &= \sum_{\ell_{ij} \in L_{\pi,S}} x_{\max(i,j)} + x_{\pi(1)} + x_{\pi(q+1)} - \sum_{i=0}^{N-1} x_i, \\ L^+_{\pi,S} &= \{\ell_{\min(i,j),\max(i,j)} : \ell_{ij} \in L_{\pi,S}\}. \end{aligned} \tag{93}$$

This depends only on the undirected tree, apex, and unordered spine endpoints. For m spine vertices, choose the vertex set in $\binom{N}{m}$ ways, its order up to reversal in $m!/2$ ways, the apex in $N-m$ ways, and each of the $N-m-1$ leaf attachments in m ways. Reversal has no fixed point since $m \geq 2$, so Equation (6) gives

$$\left|\{(h^+_{\pi,S}, L^+_{\pi,S}) : (\pi, S) \text{ a caterpillar description}\}\right| \leq C_N. \tag{94}$$

Reversing an edge swaps its omitted and linear states. Thus choosing states before or after normalization gives the same family $\mathcal{K}_0$ of descriptions, with $|\mathcal{K}_0| \leq 3^{N-2}C_N$. Write $U_k := U_{h,M}$ for $k = (h, M)$. We merge equal descriptions; distinct descriptions may still represent the same function.

Among them are the descriptions $(\ell_{ij}, \varnothing)$ representing $x \mapsto \sigma(x_i - x_j)$, with $i \neq j$: for the all-cut path $S = \{1, \ldots, r\}$ with apex $\pi(0) = j$ and last spine vertex $\pi(r+1) = i$, omitting every edge function gives

$$U^0_{\pi,S} = \sigma \circ \ell_{ij} \qquad (0 \leq i \neq j < N). \tag{95}$$

Let $\mathcal{P} \subseteq \mathcal{K}_0$ consist of descriptions with empty hinge set and head a coordinate difference. The distinct heads ℓ_{ij} give $|\mathcal{P}| \geq N(N-1)$. Put

$$\mathcal{K}_{\text{ess}} := \mathcal{K}_0 \setminus \mathcal{P}. \tag{96}$$

Every face function in $\mathcal{D}$ equals U_k for its normalized description $k \in \mathcal{K}_0$, and by Equation (8) every unit indexed by $\mathcal{P}$ lies in the auxiliary span A_N. Hence the units U_k, $k \in \mathcal{K}_{\text{ess}}$, span V_N modulo A_N. The selected basis descriptions lie in this set, so

$$q_N = |K| \leq |\mathcal{K}_{\text{ess}}| \leq 3^{N-2}C_N - N(N-1). \tag{97}$$

Step 1.2 (Compile the network and establish its widths). The network of Lemma 7 has rational weights, zero biases, first hidden layer width h_1, and second hidden layer width

$$q_N + 2 \leq 3^{N-2}C_N - N(N-1) + 2 = h_2. \tag{98}$$

Zero padding realizes the stated widths exactly.

For the growth rates, count the ordered choices of m spine labels:

$$\binom{N}{m}\frac{m!}{2} = \frac{N!}{2(N-m)!} \leq N^m \qquad (2 \leq m \leq N-1). \tag{99}$$

Using $N - m \leq N$ and $m^{N-m-1} \leq N^{N-m-1}$ in each summand of Equation (6) gives

$$\begin{aligned} C_N &\leq \sum_{m=2}^{N-1} N^m \, N \, N^{N-m-1} = (N-2)N^N \\ &\leq N^{N+1} \leq N^{4N}. \end{aligned} \tag{100}$$

Since $3^{N-2} \leq N^N$ for $N \geq 3$,

$$h_2 \leq 3^{N-2}(N-2)N^N + 2 \leq 2N^{5N}, \tag{101}$$

$$h_1 \leq 4N^2. \tag{102}$$

Taking logarithms gives the asymptotic bounds in Theorem 1. □

C Proofs of the lower bounds

C.1 Proof of the exponential count bound

A shallow function is a linear combination of linear functions and functions $x \mapsto \sigma(\ell(x))$ with ℓ linear; these form $\mathcal{L}_1$.

Lemma 14 (A common test at a tied maximum)**.** *Fix a finite family of units of the form Equation* (36)*, a set T of $m \geq 3$ labels, and a strict order of the other labels. There is a point p with $p_t = 1$ for $t \in T$ and distinct coordinates below* 1 *outside T, in the prescribed order, with the following property. Let ι_T extend vectors by zero outside T, fix $t_0 \in T$, and define*

$$\mathcal{J}_{p,T} f := (y \mapsto \partial_p f(\iota_T y))|_{y_{t_0}=0} \pmod{\mathcal{L}_{m-2}}. \tag{103}$$

This linear test annihilates every shallow function and is nonzero on $\max_N$*. For $U = \sigma(A)$, a nonzero test requires that A vanish identically on this ordered tie stratum. Its tied germ then has the form $\sigma(H(y) + \sum_{i,j \in T} c_{ij}\sigma(y_i - y_j))$, with $H(\mathbf{1}_T) = 0$, and the induced comparison graph on T has at most two connected components. An isolated tied label must have nonzero coefficient in H.*

Proof. Each preactivation restricts to an affine function of the outside coordinates on the stratum: all comparison signs are fixed. Choose p outside the zero sets of the restrictions that are not identically zero. There are finitely many such proper affine hyperplanes. If $A(p) \neq 0$, the germ of U is zero or shallow. Otherwise the restriction vanishes identically, so every outside coefficient and the constant term vanish. The latter is $H(\mathbf{1}_T)$, since the tied coordinates equal 1.

Directional germs and linear restriction preserve shallow functions. The germ of the maximum is $\max_{i \in T} y_i$; fixing $y_{t_0} = 0$ gives the support function of an $(m-1)$-simplex, which is not in $\mathcal{L}_{m-2}$. Indeed, successive opposite-face differences and facet restrictions, using Equation (43), would otherwise put a segment support function in $\mathcal{L}_0$, although it is not linear.

If the induced graph has c components, its comparison forms span $m - c$ directions. Adjoining H gives at most $m - c + 1$ directions. This bounds the support-function dimension even for signed coefficients: split the preactivation as $h_P - h_Q$ in that span and use $\sigma(h_P - h_Q) = h_{\mathrm{conv}(P \cup Q)} - h_Q$. Thus $c \geq 3$ makes the test zero. If an isolated tied label has zero coefficient in H, all forms annihilate both its coordinate vector and $\mathbf{1}_T$. These independent vectors leave at most $m - 2$ directions, including when that label is the anchor t_0. □

Proof of Theorem 9. We prove Theorem 11 and specialize to caterpillars. For a unit-inner-weight forest function, put $\kappa_o = h_o + |\{j : (o, j) \in M\}|$. At an ordered top tie T, let $u(o)$ count the neighbours of o with higher coordinate. Collecting the frozen coefficients gives $\kappa_o - u(o)$ for every $o \notin T$; an isolated tied label has head coefficient κ_t. By Lemma 14, detection requires

$$u(o) = \kappa_o \qquad (o \notin T), \tag{104}$$

$$\kappa_t \neq 0 \qquad (t \in T \text{ with no neighbour in } T). \tag{105}$$

Call a unit *passing* when these conditions hold. Every ordered tie of size at least three is passed by some unit in the expansion.

A tie set is determined by the reduced forest. Call a degree-one vertex with $\kappa = 0$ a *pendant*, and let Φ contain the nonisolated vertices that are not pendants. For a passed tie set T, an original isolate is tied exactly when its κ is nonzero. A pendant is tied exactly when its neighbour is: either mismatch violates Equations (104) and (105). A two-pendant component is entirely tied, since otherwise its lower vertex violates balance. Thus T is determined by $A = T \cap \Phi$, across all cardinalities, and $|T| \geq |A|$.

For fixed T, balance prescribes the out-degrees of $F - T$ when edges point towards the higher endpoint. A forest orientation is uniquely determined by its out-degrees: orient the edge at a leaf, delete the leaf, and repeat. Let $a_U(T)$ be the fraction of outside orders passed by U. For $0 < y \le 1$, set $F_U(y) = \sum_T a_U(T) y^{|T|}$. Averaging the covering inequality over outside orders and summing over all tied sizes gives

$$\sum_{m=3}^{N} \binom{N}{m} y^m \le \sum_U F_U(y). \tag{106}$$

It remains to bound F_U by products on disjoint blocks of Φ. The injection $T \mapsto A$ and $y^{|T|} \le y^{|A|}$ allow each subset of each block to be counted once. Relative orders in disjoint blocks are independent.

Edges and three-vertex paths. When the reduced degree is at most two, partition each nontrivial path into edges and, if its size is odd, one three-vertex path. An isolated vertex of Φ has a pendant neighbour; fix one. These fixed neighbours are distinct and outside Φ. If the singleton is outside T, its fixed edge has a prescribed order, giving the factor $1/2 + y$. For an edge and a three-vertex path, respectively, the factors are

$$E(y) = \tfrac{1}{2} + 2y + y^2, \qquad P(y) = \tfrac{1}{3} + 2y + 3y^2 + y^3. \tag{107}$$

For the empty subset of a three-vertex path, a prescribed orientation allows at most two of the six orders. Selecting only the middle vertex leaves no comparison, whereas selecting only a leaf leaves one prescribed edge; their combined contribution is $2y$. All other coefficients count subsets without further restrictions. At $y_0 = \sqrt{3/2} - 1$, we have $E(y_0) = 1$, $P(y_0) = 5/6 + y_0/2 < 1$, and $1/2 + y_0 < 1$. Thus $F_U(y_0) \le 1$, and Equation (106) proves Equation (34) and the degree-two case of Equation (35).

General reduced degree. An inclusion-minimal edge cover of the nonisolated reduced vertices is a disjoint union of stars: it contains no three-edge path, whose middle edge could be removed. Each star has at most Δ leaves. Whenever its centre and a leaf are both outside A, retain one edge; its prescribed order costs a factor $1/2$. A star with s leaves therefore has factor

$$W_s(y) = (\tfrac{1}{2} + y)(1 + y)^s + \tfrac{1}{2} y^s. \tag{108}$$

Use $y = (2\Delta + 2)^{-1} \le 1/6$. Singletons and one-leaf stars have factors at most $2/3$ and $31/36$. For $2 \le s \le \Delta$, $W_s(y) \le (1/2 + y) e^{1/2 - y} + y^2/2 \le (2/3) e^{1/3} + 1/72 < 1$. Hence again $F_U(y) \le 1$, proving Equation (35). Caterpillar face functions have reduced degree at most two, as required. □

Arbitrary tree shapes with a source orientation. The same path blocks work whenever a covering subforest on Φ has maximum degree two and each isolated vertex either has a pendant neighbour or is forced into every passed tie set. A forced singleton has factor y. For $U_{a,s,r,T}$, only the source s and head root r can be *out-leaves*, meaning leaves with $\kappa = 1$. Root the tree at r. Include every out-leaf edge, then choose one child edge at each nonleaf having neither a pendant nor an out-leaf neighbour. These edges lie in the reduced forest. Each vertex has at most one selected child edge: only s can be an out-leaf child, an out-leaf root has just one child, and a vertex choosing a child had no selected child edge. It also has at most one parent edge. The covering forest therefore has maximum degree two. Its isolated vertices in Φ either have a pendant neighbour or are leaves with $\kappa \notin \{0, 1\}$, which balance forces into T. Thus $F_U(y_0) \le 1$ for this family too, proving the asserted extension.

Large matchings in the same counting argument. First remove biases, if present, by taking the recession function $\lim_{t\to\infty} F(tx)/t$. This preserves the target, comparison graphs, and number of terms. For an added weighted graph G, now let $a_U(T)$ be the indicator that $G[T]$ has at most two components. Together with the base-family passing fractions, these capacities still satisfy Equation (106), by the common test.

Choose T by selecting each label independently with probability p. Orient a pseudoforest with in-degree at most one, independently of its ReLU orientations. For each of k disjoint matching edges, the event that its parent is absent and its child present gives a vertex with in-degree zero in $G[T]$. Each induced component has at most one such vertex. These independent events each have probability $p(1-p)$, so

$$\Pr\{c(G[T]) \le 2\} \le \rho^k \left(1 + ka + \binom{k}{2} a^2\right), \qquad \rho = 1 - p(1-p), \quad a = \frac{p(1-p)}{\rho}. \tag{109}$$

Set $p = 1/3$, equivalently $y = p/(1-p) = 1/2$, so $\rho = 7/9$ and $a = 2/7$. The added units cover probability at most $\rho^k d_k$. For the base units put $\tau = \sqrt{7}/2$. The singleton factors at $y = 1/2$ are at most τ, while $E(1/2) = 7/4 = \tau^2$ and $P(1/2) = 53/24 \le \tau^3$. Hence $F_U(1/2) \le \tau^{|\Phi|} \le \tau^N$, and a base unit covers probability at most $(2/3)^N \tau^N = \rho^{N/2} \le \rho^k$. Multiplying Equation (106) by $(2/3)^N$ consequently yields

$$\Pr\{\mathrm{Bin}(N, 1/3) \ge 3\} \le Z \rho^k d_k. \tag{110}$$

The left side is increasing in N and exceeds $1/2$ at $N = 9$. This proves Equation (37).

For completeness, define the excess of a graph by $b(G) = \sum_C \max(0, |E(C)| - |V(C)|)$ over its connected components. Extend a maximum matching to a spanning forest and add one omitted edge in each originally cyclic component. The resulting pseudoforest preserves the matching and omits exactly $b(G)$ edges. Restoring those edges reduces the induced component count by at most $b(G)$. For excess at most b, replace d_k by $\sum_{j=0}^{\min(k,b+2)} \binom{k}{j}(2/7)^j$. When $k = \Omega(N)$ and $b = o(N)$, this sum is $\exp(o(N))$. The leaf criterion follows from a minimum vertex cover C of a tree, whose size equals its maximum matching number ν. The nonleaf vertices outside C form an independent set I. Their at least $2|I|$ incident edges lie in the forest on $I \cup C$, so $2|I| \le |I| + \nu - 1$ and $n - L \le 2\nu - 1$.

All tree orientations. For Corollary 12, apply the common test to triples. Balance forces the apex into the tie, since its coefficient is -1 and no comparison edge meets it. The apex is isolated, so the other two tied labels must be the endpoints of a retained edge. Each unit has at most $N - 2$ such edges. Covering all triples gives $\binom{N}{3} \le (N-2)|K|$, proving the claim.

Adding every oriented star. Work in full coordinates. For a label j, let ι_j insert a zero at that coordinate and put $\mathcal{T}_j g = (\partial_{-e_j} g - \partial_{e_j} g) \circ \iota_j$. This linear operator sends $\mathcal{L}_d$ into $\mathcal{L}_{d-1}$ for $d \ge 1$ and annihilates $\mathcal{L}_0$: if coordinate j is constant on a polytope, its two exposed faces coincide; otherwise both have smaller dimension. Projection under restriction cannot increase dimension. Moreover, $\mathcal{T}_j \max_m = \max_{m-1}$, since the simplex faces are its facet opposite e_j and the vertex e_j.

For a canonical star, deletion of a nonroot leaf gives the smaller star modulo $\mathcal{L}_1$, with sign $+$ for $\sigma(x_l - x_c)$ and sign $-$ for $\sigma(x_c - x_l)$. If the deleted leaf is the head root, the latter orientation transfers the head to the centre with sign $+$; the former gives zero modulo $\mathcal{L}_1$. Deleting the apex leaves a zonotope, and deleting the centre fixes every segment, so these cases also give zero modulo $\mathcal{L}_1$. Thus successive deletions preserve the span of canonical stars modulo the lower-dimensional terms.

At five labels, a six-flag certificate gives a linear functional that annihilates every canonical oriented star and $\mathcal{L}_3$, and takes value 120 on the simplex. The independent vertex checker checks all 640 labelled stars, with every head root and orientation, by successive maxima and minima over explicit vertex sets. Composing this functional with $m - 5$ coordinate deletions therefore gives such a separator at every $m \ge 5$. A nonhinge state already has dimension at most $m - 2$, so all star face states are covered too.

Assign every term whose function belongs to the star family to the added star class, even if it was originally listed in another class. At any tie of size at least five, a surviving top-dimensional star germ is again a canonical star: each untied component has a unique attachment to the tied subtree, and summing its zero frozen coefficients transports the head to that attachment if it contains the original root. Apply the separator to the local expansion. The star terms vanish, so some nonstar term must cover the locality. Thus Equation (106) holds with the sum starting at $m = 5$ and only nonstar terms on the right. At y_0 this gives the same exponential rate, subtracting only a polynomial of degree four. For the matching mixture, replace the tail in Equation (110) by $\Pr\{\mathrm{Bin}(N, 1/3) \geq 5\}$, which exceeds $1/2$ for $N \geq 15$. This proves the claims without counting the added stars.

C.2 Proof of the dimension bound

Proof of Theorem 10. Put $r = N - 2$ and $b = N - 1$. For every permutation π fixing 0 and b, take the path $\pi(1), \ldots, \pi(r), b$, with apex 0 and every edge a hinge:

$$U_\pi(x) = \sigma\Big(x_b - x_0 + \sum_{j=1}^{r} \sigma(x_{\pi(j)} - x_{\pi(j+1)})\Big) \in V_N. \tag{111}$$

For an order ρ, put $c = \rho(r)$ and choose

$$p^\rho_0 = p^\rho_b = 1, \qquad p^\rho_{\rho(j)} = 1 + j - r \quad (1 \leq j \leq r). \tag{112}$$

The three labels $0, b, c$ tie at the top. Use the test $\mathcal{J}_\rho = \mathcal{J}_{p^\rho, \{0,b,c\}}$ from Equation (103), anchored at 0. It annihilates A_N and indeed every shallow function. For $\pi = \rho$, only the last edge can activate in the tied germ, which is $\sigma(y_b - y_0 + \sigma(y_c - y_b)) = \max(y_0, y_b, y_c) - y_0$. Its anchored class is the nonzero triangle class $v = [\max(0, s, t)]$ modulo $\mathcal{L}_1$. For $\pi \neq \rho$, the path order has a strict descent in ρ-height, so its outer preactivation is strictly positive at p^ρ. Its germ is therefore shallow. Identifying the two free tied coordinates with (s, t) gives

$$\mathcal{J}_\rho U_\pi = \begin{cases} v, & \pi = \rho, \\ 0, & \pi \neq \rho. \end{cases} \tag{113}$$

Applying $\mathcal{J}_\rho$ to any relation $\sum_\pi c_\pi U_\pi \in A_N$ gives $c_\rho v = 0$, hence $c_\rho = 0$. The $r!$ paths are independent modulo A_N, proving $q_N \geq (N - 2)!$. □

D Relation to the construction of Brunck et al.

The combinatorial proof of [BHR+26, Part 2] cones zonotopes $B_{v,\pi,M}$ to pyramids and expresses the simplex support function as a signed sum of their support functions modulo lower-dimensional terms [BHR+26, Lemma 22]. This appendix identifies these building blocks. For a nonempty set M of marked positions, the pyramid has the support function of a translated caterpillar polytope, with apex v, label order π, and cut set $M - 1$. For $M = \varnothing$ it gives a rooted star, which is not a caterpillar function. We use the indexing of [BHR+26, Section 7.1], with their n written as N and $[N]$ identified with our label set $\{0, \ldots, N - 1\}$. Positions of an ordering are numbered from 1.

Definition 15 (Zonotopes of Brunck et al., Lean_23, Lean_24)**.** Let $N \geq 3$, let v be a label, let $\pi = (\pi_1, \dots, \pi_{N-1})$ be an ordering of the remaining labels, and let $M \subseteq \{2, \dots, N-1\}$. For $p \in \mathbb{R}^N$ put $u_i(p) := \sum_{1 \leq j < i} p_{\pi_j}$. Following [BHR+26, Equation (3)], define

$$H_{-v} := \Big\{ p \in \mathbb{R}^N : p_v = 0,\ \sum_k p_k = 1 \Big\}, \tag{114}$$

$$B_{v,\pi,M} := \left\{ p \in H_{-v} : \begin{array}{ll} 0 \leq u_i(p) \leq 1 & \text{for } i \in M, \\ -1 \leq p_{\pi_i} \leq 0 & \text{for } i \in \{2, \dots, N-1\} \setminus M \end{array} \right\}. \tag{115}$$

The pyramid with apex e_v is $\mathrm{pyr}_{e_v}(B_{v,\pi,M}) := \mathrm{conv}(\{e_v\} \cup B_{v,\pi,M})$, with support function $h_{v,\pi,M}$ as in Equation (16). Its last marked position and the marked parent of a position i are

$$q := \max(\{1\} \cup M), \qquad r(i) := \max\big(\{1\} \cup \{j \in M : j < i\}\big) \qquad (2 \leq i \leq N-1). \tag{116}$$

For $M \neq \varnothing$, $q = \max M$. For labels $a \neq c$, the *rooted star* with apex a and root c is

$$S_{a,c}(x) := \sigma\Big(x_c - x_a + \sum_{l \notin \{a,c\}} \sigma(x_c - x_l) \Big). \tag{117}$$

Theorem 16 (Relation to Brunck et al., Lean_25, Lean_26, Lean_27, Lean_28, Lean_29, Lean_30, Lean_31, Lean_32, Proof)**.** *Let v, π, and M be as in Definition 15, and let q and $r(i)$ be as in Equation* (116)*.*

(a) The base is the zonotope

$$B_{v,\pi,M} = e_{\pi_q} + \sum_{i=2}^{N-1} \big[0, e_{\pi_{r(i)}} - e_{\pi_i}\big]. \tag{118}$$

(b) For every $x \in \mathbb{R}^N$,

$$h_{v,\pi,M}(x) = x_v + \sigma\Big(x_{\pi_q} - x_v + \sum_{i=2}^{N-1} \sigma\big(x_{\pi_{r(i)}} - x_{\pi_i}\big) \Big). \tag{119}$$

(c) Let $M \neq \varnothing$. Define the caterpillar description (π', S) by $\pi'(0) := v$, $\pi'(j) := \pi_j$ for $1 \leq j \leq N-1$, and $S := \{i - 1 : i \in M\}$. Then $|S| = |M|$ and

$$h_{v,\pi,M}(x) = x_v + U_{\pi',S}(x) \qquad (x \in \mathbb{R}^N). \tag{120}$$

Hence $h_{v,\pi,M} - x_v = U_k$ for some $k \in \mathcal{K}_0$. Conversely, every caterpillar description (π', S) arises in this way from $v = \pi'(0)$, $\pi_j = \pi'(j)$, and $M = \{j + 1 : j \in S\} \neq \varnothing$.

(d) Let $M = \varnothing$. Then $h_{v,\pi,\varnothing}(x) = x_v + S_{v,\pi_1}(x)$. No rooted star is a caterpillar function: $S_{a,c} \neq U_k$ for all labels $a \neq c$ and all $k \in \mathcal{K}_0$. In particular, $h_{v,\pi,\varnothing} - x_v \neq U_k$ for every $k \in \mathcal{K}_0$.

By Equation (17), the right-hand side of Equation (120) is the support function of the translated caterpillar polytope $e_v + Q_{h_{\pi',S}, L_{\pi',S}}$. The family of [BHR+26] therefore consists of the caterpillar polytopes, translated by e_v, and of the pyramids over rooted stars. The sign $(-1)^{|M|}$ of [BHR+26, Lemma 22] equals $(-1)^{|S|}$. The theorem compares building blocks only. It makes no statement about corrections or network coefficients.

Proof of Theorem 16. Write $I := \{2, \dots, N-1\}$ and $g_i := e_{\pi_{r(i)}} - e_{\pi_i}$ for $i \in I$. Every element of $\{1\} \cup M$ lies in $\{1, \dots, N-1\}$, so q and all $r(i)$ are positions. Since $M \subseteq I$, the definitions give

- $1 \le r(i) < i$ for $i \in I$, and $i \le r(l)$ whenever $i \in M$ and $i < l$,
- $i \le q$ for $i \in M$,
- $q \ne i$ and $r(l) \ne i$ for $i \in I \setminus M$ and all l.

Step 16.1 (Coordinates of the zonotope). For $t \in [0,1]^I$ let $z(t) := e_{\pi_q} + \sum_{l \in I} t_l g_l$. Every e_{π_j} has coordinate sum 1 and vanishes at v, so $z(t) \in H_{-v}$. For $i \in I \setminus M$ the coordinate $z(t)_{\pi_i}$ collects the term $-t_i$ and no other term, since π_i is neither π_q nor any $\pi_{r(l)}$. Hence $z(t)_{\pi_i} = -t_i$. For $i \in M$, the prefix sum u_i counts a vector e_{π_j} exactly when $j < i$. Thus e_{π_q} contributes 0, because $q \ge i$. A generator g_l with $l < i$ contributes $1 - 1 = 0$, because $r(l) < l < i$. A generator with $l > i$ contributes $0 - 0$, because $r(l) \ge i$. Finally g_i contributes t_i, because $r(i) < i$. Hence $u_i(z(t)) = t_i$. Therefore $z(t) \in B_{v,\pi,M}$.

Step 16.2 (The coordinates determine the point). Let $p, p' \in H_{-v}$ with $u_i(p) = u_i(p')$ for $i \in M$ and $p_{\pi_i} = p'_{\pi_i}$ for $i \in I \setminus M$. Put $d := p - p'$ and $U_i := \sum_{1 \le j < i} d_{\pi_j}$. Then $U_1 = 0$, and $U_N = \sum_k d_k - d_v = 0$. Descending from $i = N-1$ to $i = 2$, either $i \in M$ and $U_i = 0$ by assumption, or $i \notin M$ and $U_i = U_{i+1} - d_{\pi_i} = U_{i+1} = 0$. Hence $d_{\pi_j} = U_{j+1} - U_j = 0$ for $1 \le j \le N-1$, and $d_v = 0$. Thus $p = p'$. This is the invertibility in [BHR+26, Claim 21].

Step 16.3 (Proof of (a)). Step 16.1 shows that the right-hand side of Equation (118) lies in $B_{v,\pi,M}$. Conversely, let $p \in B_{v,\pi,M}$, and set $t_i := u_i(p)$ for $i \in M$ and $t_i := -p_{\pi_i}$ for $i \in I \setminus M$. The inequalities in Equation (115) say $t \in [0,1]^I$. By Step 16.1, the points p and $z(t)$ of H_{-v} have the same coordinates, so $p = z(t)$ by Step 16.2.

Step 16.4 (Proof of (b)). A linear function attains the same supremum on a set and on its convex hull. Hence $h_{v,\pi,M}(x) = \max(x_v, \sup_{p \in B_{v,\pi,M}} \langle p, x \rangle)$. Put $b_l := x_{\pi_{r(l)}} - x_{\pi_l}$. By (a), $\langle z(t), x \rangle = x_{\pi_q} + \sum_l t_l b_l$, and $t_l b_l \le \sigma(b_l)$ for $0 \le t_l \le 1$, with equality for $t_l = \mathbb{1}_{\{b_l \ge 0\}}$. So the supremum over $B_{v,\pi,M}$ equals $A := x_{\pi_q} + \sum_l \sigma(b_l)$. Then Equation (7) gives $\max(x_v, A) = x_v + \sigma(A - x_v)$, which is Equation (119).

Step 16.5 (Proof of (c)). With π' and S as in (c), position j of the caterpillar carries the label $\pi'(j+1) = \pi_{j+1}$, so caterpillar position j is the position $j+1$ of [BHR+26]. The map $i \mapsto i - 1$ is a bijection from M onto S, hence $|S| = |M|$, and it carries $\{1\} \cup M$ onto $\{0\} \cup S$ and $\{1\} \cup \{j \in M : j < i\}$ onto $\{0\} \cup \{j \in S : j < i - 1\}$. Comparing with Equation (10), the last spine position is $q - 1$ and the parent of caterpillar position $i - 1$ is $r(i) - 1$. By Equations (12) and (13), the head is $x_{\pi_q} - x_v$ and the edge functions are $x_{\pi_{r(i)}} - x_{\pi_i}$, $i \in I$. So the right-hand side of Equation (119) is $x_v + U_{\pi',S}(x)$. If $M \ne \varnothing$, then $S \ne \varnothing$ and $0 \notin S$, so (π', S) is a caterpillar description, and its normalized description $k \in \mathcal{K}_0$ satisfies $U_k = U_{\pi',S}$ by Equation (8). The converse inverts these bijections.

Step 16.6 (Proof of (d)). For $M = \varnothing$, $q = 1$ and $r \equiv 1$, and $\{\pi_i : i \in I\}$ is the set of labels other than v and π_1. So Equation (119) becomes $x_v + S_{v,\pi_1}(x)$. Let $a \ne c$ and evaluate at $x = e_c$. Every inner term of $S_{a,c}(e_c)$ equals $\sigma(1)$, so $S_{a,c}(e_c) = 1 + (N-2) = N - 1$. Let $k \in \mathcal{K}_0$. Then U_k is the normalization of a face function $U^{\tau}_{\pi',S}$, and $U_k = U^{\tau}_{\pi',S}$ by Equation (8). Since $\ell \le \sigma \circ \ell$, $0 \le \sigma \circ \ell$, and σ is monotone, $U^{\tau}_{\pi',S} \le U_{\pi',S}$. At $x = e_c$, every head and edge function takes values in $\{-1, 0, 1\}$. If $c = \pi'(0)$, the argument of the outer σ in $U_{\pi',S}(e_c)$ is -1, so $U_{\pi',S}(e_c) = 0$. Otherwise c is the label of a caterpillar position m. The head contributes at most 1, and every inner hinge at most 1. If $m \ge 1$, the inner hinge of the edge into position m is $\sigma(x_{\pi'(p(m)+1)} - 1) = 0$, so the value is at most $1 + (N-3)$. If $m = 0$, the head contributes 0, because $q \ge \min S > 0$, so the value is at most $N - 2$. Thus $U_k(e_c) \le N - 2 < S_{a,c}(e_c)$. □

E Lean definitions and main theorem

The real parameter arrays in Equation (4) have dimensions

$$\begin{aligned} W_1 &\in \mathbb{R}^{h_1 \times N}, \quad b_1 \in \mathbb{R}^{h_1}, \\ W_2 &\in \mathbb{R}^{h_2 \times h_1}, \quad b_2 \in \mathbb{R}^{h_2}, \\ w_3 &\in \mathbb{R}^{h_2}, \qquad b_3 \in \mathbb{R}, \end{aligned} \tag{121}$$

Sources (`Definitions.lean`, `Theorems.lean`). The listings specify the network, widths, and correctness for every real input. They reproduce declarations from these files at revision `8b4d74ab66f4`, omitting imports, documentation comments, and the theorem's proof body; these remain in the linked sources.

E.1 Network and width definitions

```
namespace MaxCaterpillar

open BigOperators

abbrev Point (N : ℕ) := Fin N → ℝ

noncomputable def relu (t : ℝ) : ℝ := max 0 t

structure Network (N h₁ h₂ : ℕ) where
  W₁ : Fin h₁ → Fin N → ℝ
  b₁ : Fin h₁ → ℝ
  W₂ : Fin h₂ → Fin h₁ → ℝ
  b₂ : Fin h₂ → ℝ
  w₃ : Fin h₂ → ℝ
  b₃ : ℝ

noncomputable def Network.eval {N h₁ h₂ : ℕ} (net : Network N h₁ h₂)
    (x : Point N) : ℝ :=
  (∑ k, net.w₃ k * relu
    ((∑ j, net.W₂ k j * relu
      ((∑ i, net.W₁ j i * x i) + net.b₁ j)) + net.b₂ k))
    + net.b₃

def Network.HasRationalWeights {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  (∀ j i, ∃ q : ℚ, (q : ℝ) = net.W₁ j i) ∧
    (∀ k j, ∃ q : ℚ, (q : ℝ) = net.W₂ k j) ∧
    ∀ k, ∃ q : ℚ, (q : ℝ) = net.w₃ k

def Network.HasZeroBiases {N h₁ h₂ : ℕ}
    (net : Network N h₁ h₂) : Prop :=
  net.b₁ = 0 ∧ net.b₂ = 0 ∧ net.b₃ = 0
```

```
noncomputable def Maximum (N : ℕ) [NeZero N] (x : Point N) : ℝ :=
  Finset.univ.sup'
    ⟨⟨0, Nat.pos_of_ne_zero (NeZero.ne N)⟩,
      Finset.mem_univ _⟩ x

def firstWidth (N : ℕ) : ℕ := N.choose 2 + N + 1

def coneCount (N : ℕ) : ℕ :=
  ∑ m in Finset.Icc 2 (N - 1),
    N.choose m * (m.factorial / 2) * (N - m)
      * m ^ (N - m - 1)

def secondWidth (N : ℕ) : ℕ :=
  coneCount N * 3 ^ (N - 2) - N * (N - 1) + 2

end MaxCaterpillar
```

E.2 Caterpillar functions and expansions

These definitions, also in `Definitions.lean`, state the results of Section 4. A description $k = (h, M)$ is stored in reduced coordinates, and `realUnitY` evaluates U_k on $\mathbb{R}^N$.

```
abbrev FaceKey (r : ℕ) :=
  MaxNN.Caterpillar.Corrections.RealKey (r + 1)

noncomputable def caterpillarDescriptions (r : ℕ) :
    Finset (FaceKey r) :=
  MaxNN.Caterpillar.Corrections.normalizedKeys r

noncomputable def caterpillarFunction {r : ℕ}
    (k : FaceKey r) : Point (r + 1 + 1) → ℝ :=
  MaxNN.Caterpillar.Corrections.realUnitY k

noncomputable def auxiliaryFunction (N : ℕ)
    (j : Fin (MaxNN.Width.W N)) : Point N → ℝ :=
  fun x => MaxNN.Width.feat N x j

def IsCaterpillarExpansion (r : ℕ)
    (K : Finset (FaceKey r)) : Prop :=
  K ⊆ caterpillarDescriptions r ∧
    ∃ (d : FaceKey r → ℝ)
      (a : Fin (MaxNN.Width.W (r + 1 + 1)) → ℝ),
      ∀ x, Maximum (r + 1 + 1) x =
        ∑ k in K, d k * caterpillarFunction k x +
          ∑ j, a j * auxiliaryFunction (r + 1 + 1) j x

noncomputable def auxiliarySpan (N : ℕ) :
    Submodule ℝ (Point N → ℝ) :=
```

```
  Submodule.span ℝ (Set.range (auxiliaryFunction N))

noncomputable def caterpillarSpan (r : ℕ) :
    Submodule ℝ (Point (r + 1 + 1) → ℝ) :=
  Submodule.span ℝ
    ((caterpillarFunction '' ↑(caterpillarDescriptions r))
      ∪ Set.range (auxiliaryFunction (r + 1 + 1)))

noncomputable def caterpillarDim (r : ℕ) : ℕ :=
  FiniteDimensional.finrank ℝ (↑(caterpillarSpan r) ⧸
    (auxiliarySpan (r + 1 + 1)).comap
      (caterpillarSpan r).subtype)
```

E.3 The main theorems

The assumption `[NeZero N]` expresses positive arity. For $N \geq 3$ the first statement is Theorem 1; the natural-number width definitions also cover $N = 1, 2$. With $N = r + 2$, the next three are Theorem 9, and the last is Theorem 10. All are in `Theorems.lean`.

```
theorem caterpillar_net (N : ℕ) [NeZero N] :
    ∃ net : Network N (firstWidth N) (secondWidth N),
      (∀ x, net.eval x = Maximum N x) ∧
        net.HasRationalWeights ∧ net.HasZeroBiases

theorem caterpillar_expansion_choose_le (r : ℕ)
    {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K)
    {k : ℕ} (hk : 3 ≤ k) (hkN : k ≤ r + 1 + 1) :
    (r + 1 + 1).choose k ≤ 6 ^ k * K.card

theorem caterpillar_expansion_exponential (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    7 ^ (r + 1 + 1) ≤
      (r + 1 + 1 + 1) * 6 ^ (r + 1 + 1) * K.card

theorem caterpillar_expansion_quadratic (r : ℕ)
    (hr : 1 ≤ r) {K : Finset (FaceKey r)}
    (hK : IsCaterpillarExpansion r K) :
    (r + 1 + 1) * (r + 1) ≤ 6 * K.card

theorem caterpillarDim_ge (r : ℕ) (hr : 1 ≤ r) :
    r.factorial ≤ caterpillarDim r
```

E.4 The building blocks of Brunck et al.

The definitions of Definition 15, in the indexing of [BHR+26, Section 7.1], are in `Competing/Defs.lean`. An ordering is a bijection from the $N - 1$ positions onto the labels other than v, and `Ordering.pos` returns

π_i at position i, $1 \le i \le N-1$.

```
noncomputable def supportFunction {n : ℕ}
    (P : Set (Fin n → ℝ)) (y : Fin n → ℝ) : ℝ :=
  sSup ((fun p => ∑ k, p k * y k) '' P)

def pyramid {n : ℕ} (q : Fin n → ℝ)
    (P : Set (Fin n → ℝ)) : Set (Fin n → ℝ) :=
  convexHull ℝ (insert q P)

def zonotope {n : ℕ} (a : Fin n → ℝ) (I : Finset ℕ)
    (g : ℕ → Fin n → ℝ) : Set (Fin n → ℝ) :=
  {p | ∃ t : ℕ → ℝ, (∀ i ∈ I, 0 ≤ t i ∧ t i ≤ 1) ∧
    p = a + ∑ i in I, t i • g i}

abbrev BHRSY.Ordering (n : ℕ) (v : Fin n) :=
  Fin (n - 1) ≃ {k : Fin n // k ≠ v}

def BHRSY.Ordering.pos {v : Fin n} (π : Ordering n v)
    (i : ℕ) : Fin n :=
  if h : i - 1 < n - 1 then (π ⟨i - 1, h⟩).1 else v

def BHRSY.hyperplane (v : Fin n) : Set (Fin n → ℝ) :=
  {p | p v = 0 ∧ ∑ k, p k = 1}

noncomputable def BHRSY.prefixSum {v : Fin n}
    (π : Ordering n v) (p : Fin n → ℝ) (i : ℕ) : ℝ :=
  ∑ j in Finset.Ico 1 i, p (π.pos j)

def BHRSY.parallelepiped (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  {p | p ∈ hyperplane v ∧
    (∀ i ∈ M, 0 ≤ prefixSum π p i ∧ prefixSum π p i ≤ 1) ∧
    ∀ i ∈ Finset.Icc 2 (n - 1), i ∉ M →
      -1 ≤ p (π.pos i) ∧ p (π.pos i) ≤ 0}

def BHRSY.paramid (v : Fin n) (π : Ordering n v)
    (M : Finset ℕ) : Set (Fin n → ℝ) :=
  pyramid (Pi.single v 1) (parallelepiped v π M)

noncomputable def BHRSY.paramidSupport (v : Fin n)
    (π : Ordering n v) (M : Finset ℕ) : (Fin n → ℝ) → ℝ :=
  supportFunction (paramid v π M)

def BHRSY.lastMarked (M : Finset ℕ) : ℕ :=
  (insert 1 M).max' (Finset.insert_nonempty _ _)

def BHRSY.markedParent (M : Finset ℕ) (i : ℕ) : ℕ :=
```

```
  (insert 1 (M.filter fun j => j < i)).max'
    (Finset.insert_nonempty _ _)
```

The dictionary uses the word $\pi' = (v, \pi_1, \ldots, \pi_{N-1})$ (`Ordering.word`), the cut set $S = M - 1$ (`cutSet`), and the description $(h_{\pi',S}, L_{\pi',S})$ of a word (`caterpillarWordDescription`). The rooted star is `BHRSY.rootedStar`. With $N = r + 2$, `Theorems.lean` states Theorem 16 as follows.

```
theorem bhrsy_parallelepiped_eq_zonotope {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    BHRSY.parallelepiped v π M =
      zonotope (Pi.single (π.pos (BHRSY.lastMarked M)) 1)
        (Finset.Icc 2 (r + 1))
        (fun i => Pi.single (π.pos (BHRSY.markedParent M i)) 1
          - Pi.single (π.pos i) 1)

theorem bhrsy_support_eq {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π M y =
      y v + MaxCaterpillar.relu
        (y (π.pos (BHRSY.lastMarked M)) - y v +
          ∑ i in Finset.Icc 2 (r + 1), MaxCaterpillar.relu
            (y (π.pos (BHRSY.markedParent M i)) - y (π.pos i)))

theorem bhrsy_support_eq_word {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) {M : Finset ℕ}
    (hM : M ⊆ Finset.Icc 2 (r + 1)) :
    (BHRSY.cutSet (r := r) M).card = M.card ∧
      ∀ y : Point (r + 1 + 1), BHRSY.paramidSupport v π M y =
        y v + caterpillarFunction
          (caterpillarWordDescription π.word (BHRSY.cutSet M)) y

theorem bhrsy_support_mem_caterpillar {r : ℕ}
    {v : Fin (r + 1 + 1)} (π : BHRSY.Ordering (r + 1 + 1) v)
    {M : Finset ℕ} (hM : M ⊆ Finset.Icc 2 (r + 1))
    (hne : M.Nonempty) :
    ∃ k ∈ caterpillarDescriptions r, ∀ y,
      BHRSY.paramidSupport v π M y = y v + caterpillarFunction k y

theorem bhrsy_support_empty {r : ℕ} {v : Fin (r + 1 + 1)}
    (π : BHRSY.Ordering (r + 1 + 1) v) (y : Point (r + 1 + 1)) :
    BHRSY.paramidSupport v π ∅ y =
      y v + BHRSY.rootedStar v (π.pos 1) y

theorem rootedStar_ne_caterpillarFunction {r : ℕ}
    {a c : Fin (r + 1 + 1)} (hac : a ≠ c) {k : FaceKey r}
    (hk : k ∈ caterpillarDescriptions r) :
    BHRSY.rootedStar a c ≠ caterpillarFunction k
```

The same file contains the converse `bhrsy_word_mem`, the special cases `bhrsy_parallelepiped_eq_zonotope_of_nonempty` and `bhrsy_parallelepiped_empty`, the statement `bhrsy_support_empty_not_caterpillar`, and the dichotomy `bhrsy_family`.

E.5 Proof routes in the formalization

The paper selects the coefficients of its single Caterpillar Net by the finite rational evaluation system in Section 3.1. The public Lean theorem `caterpillar_net` certifies rational coefficient existence and the stated width using an internal $\mathbb{Q}$-linear projection argument. The evaluation-matrix proof itself is not a new formalized declaration.

The formalized count proof uses star blocks and gives the weaker bound $(7/6)^N/(N+1)$. The written proof in Appendix C.1 uses one generic germ and sums over all tie sizes, sharpening the estimate and allowing real heads with unit inner coefficients. The formalized vocabulary-dimension proof uses a third mixed difference; Appendix C.2 gives a shorter proof using the same germ as the count argument.

The keyword `noncomputable` allows classical definitions of real quantities. The theorem is an exact representation statement; it does not by itself assert an efficient algorithm for extracting numerical weights.

E.6 Verification scope

The repository's historical verification certificate, dated 24 October 2026, records a build with Lean 4.4.0 and checks 122 public statements against their exact types. Its permitted axioms are `propext`, `Classical.choice`, and `Quot.sound`. The checked interface includes dictionary sufficiency, rational expansions, network correctness with rational weights and zero biases, the width bounds, the compilation and compression theorems, the Wang–Sun and Koutschan–Moser–Ponomarchuk–Schicho representations, and the network theorems for sums of maxima and CPWL functions with their rationality and bias statements. The equation and theorem links identify supporting declarations for the mathematical arguments in the paper. The certificate and its source hashes are available in the linked verification record. The subsequent consolidation of the Lean modules is not covered by that certificate: the current sources require an updated exact-statement manifest and a fresh verification run. In particular, the rational evaluation system, the common local test, the sharpened count bounds, and the source-orientation, matching-number, and arbitrary-star extensions are mathematical arguments in this paper, not claims checked by the historical certificate. The mathematical arguments are provided in the main text and Appendices A to C; the formalization is an additional certificate of the formal statements.

AI methodology statement

Claude Code and Codex were used for mathematical exploration and proof search, development of the Lean formalization, and drafting and revision of the manuscript. Their use materially affected the construction and proofs in Sections 3 to 7 and Appendices A to E. The author set the research objectives, directed the searches, and revised the mathematical presentation. The accompanying Lean certificate checks the formal statements and their axiom dependencies; it does not certify the prose exposition, literature comparison, or claims of originality. The author is responsible for the contents of the paper.

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